



Understanding genomes by engineering structural variation at scale

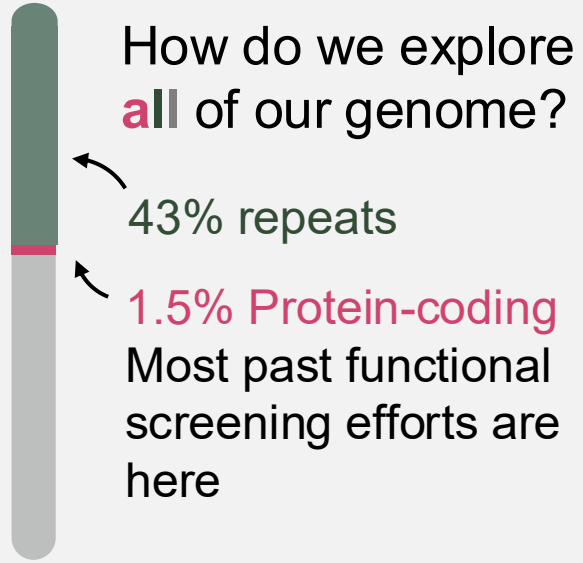
Jonas Koeppel

EMBL seminar

December 4, 2025

Koeppel group: Engineering structural variation to understand human genome function

Questions



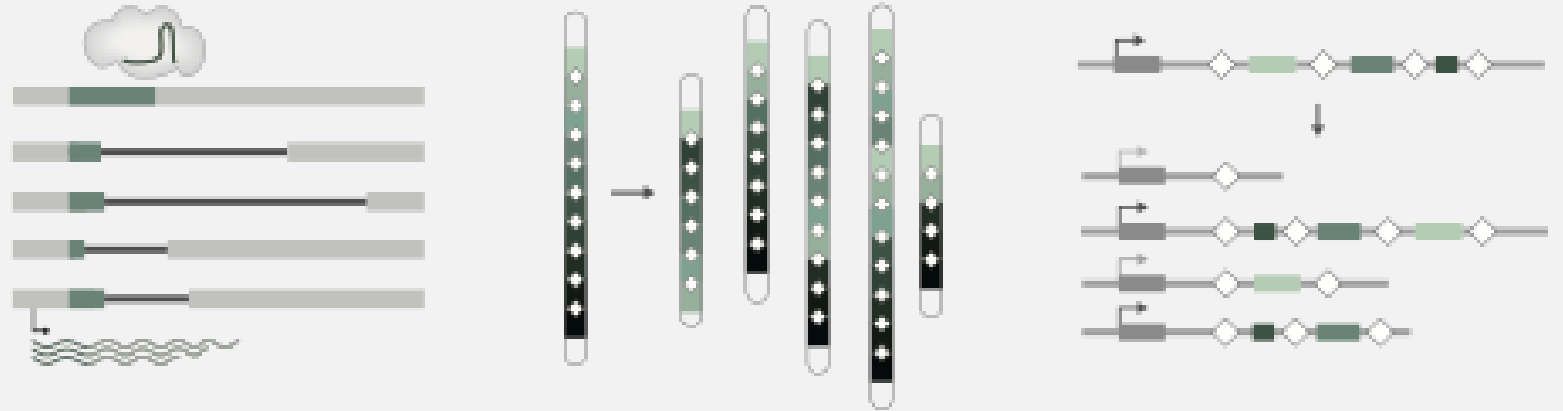
- How much is dispensable?
- How to build a minimal genome?
- Functional consequences?



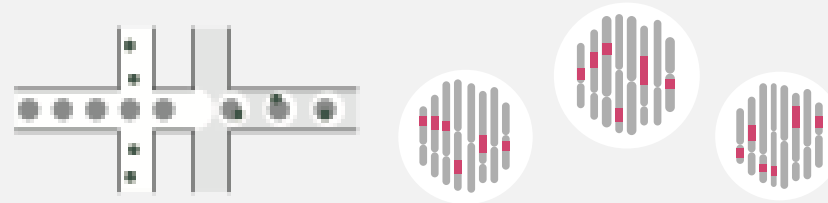
Jonas Koeppel

Approaches

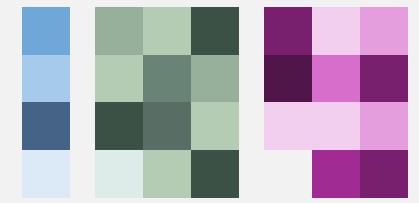
Large-scale genome engineering



Single-cells as multiplexed experiments



Machine learning



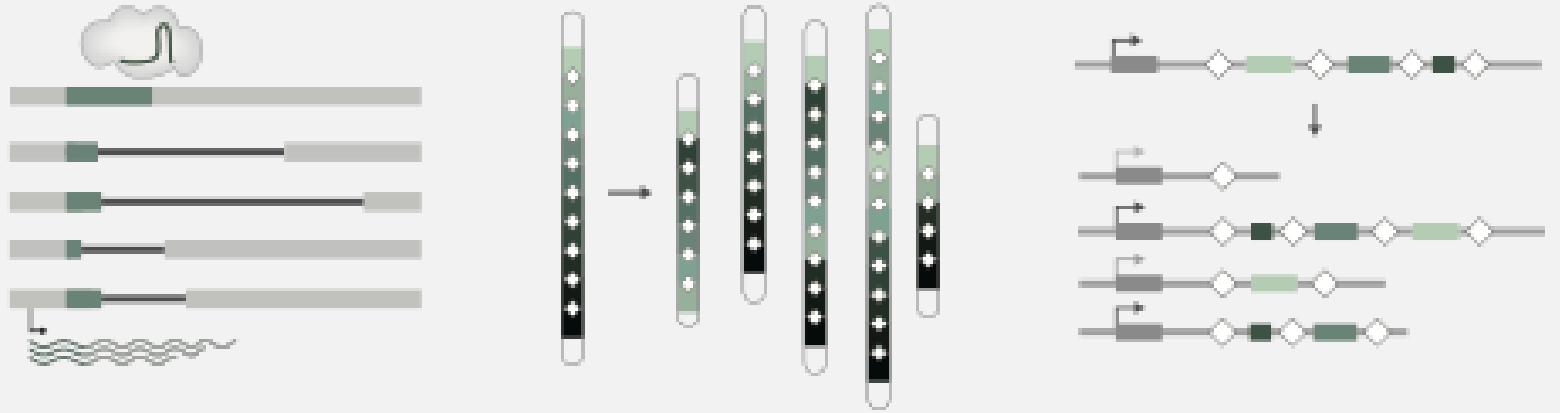
Questions



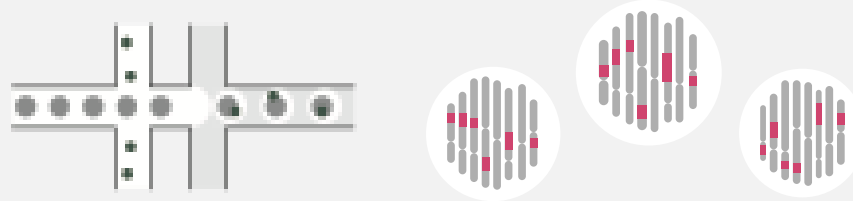
- How much is dispensable?
- How to build a minimal genome?
- Functional consequences of SVs?

Approach

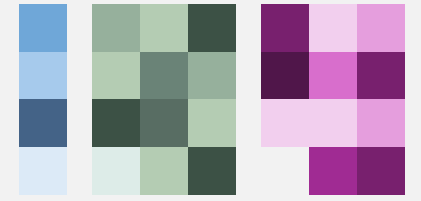
Large-scale genome engineering



Single-cells as multiplexed experiments



Machine learning

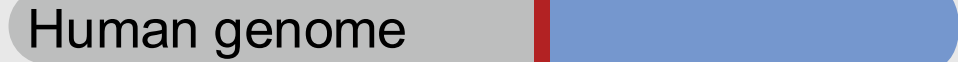


Koeppel group: Engineering structural variation to understand human genome function

Questions

What fraction of the genome is essential?

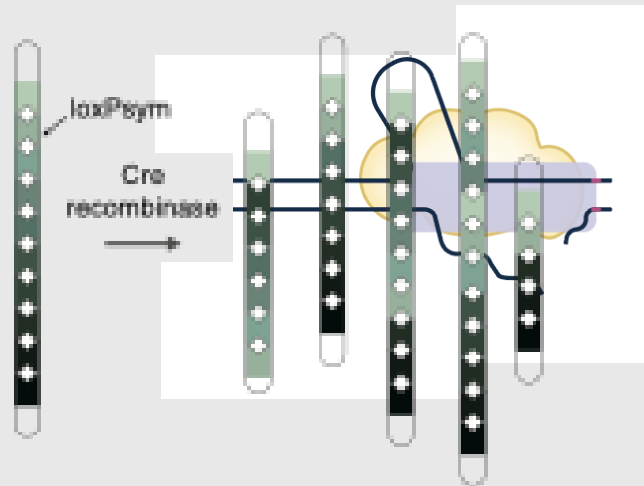
 Fugu fish genome

 Human genome

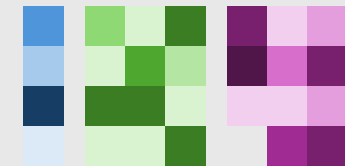


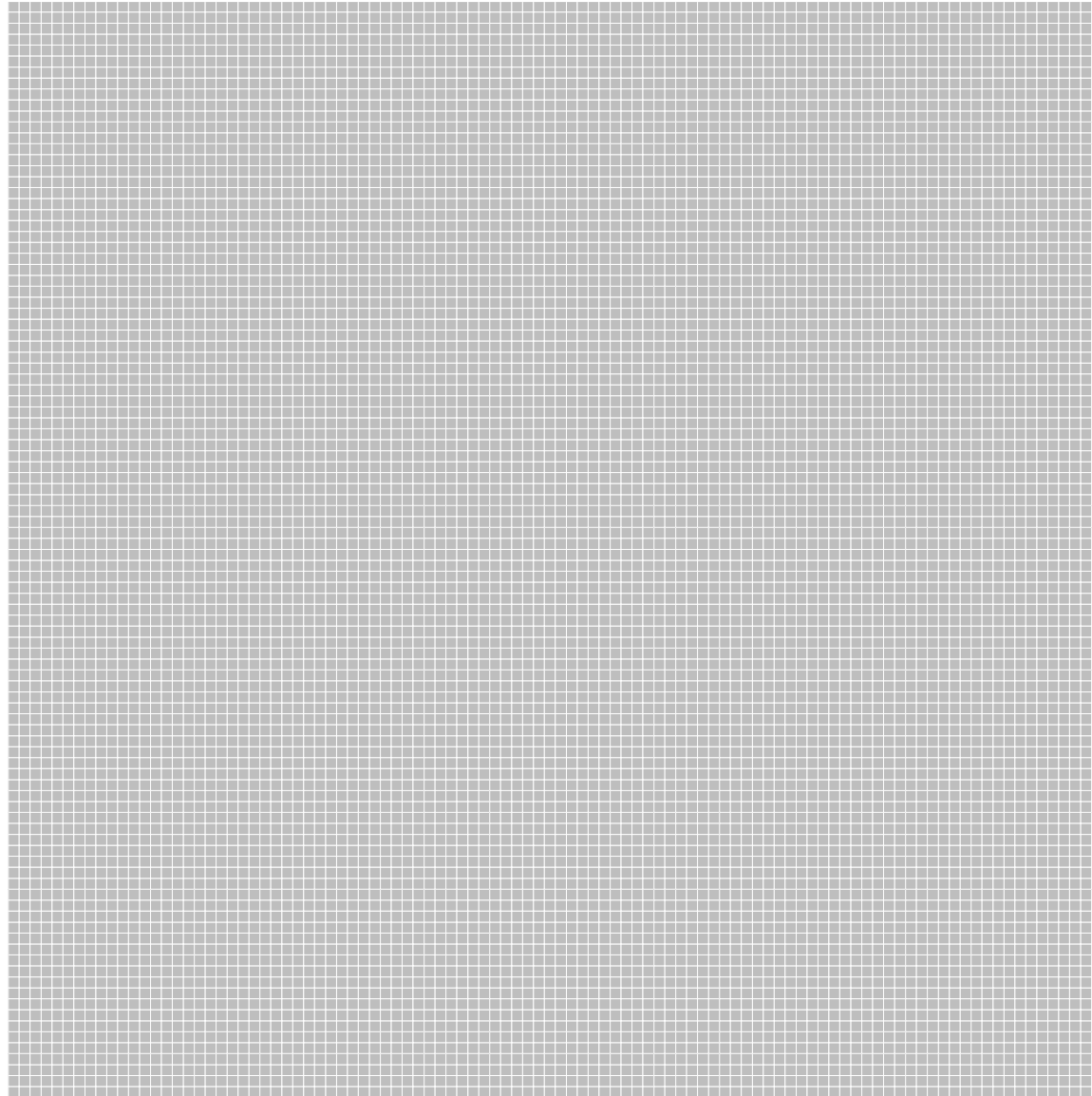
Approaches

Large-scale genome engineering



Single-cells as multiplexed experiments

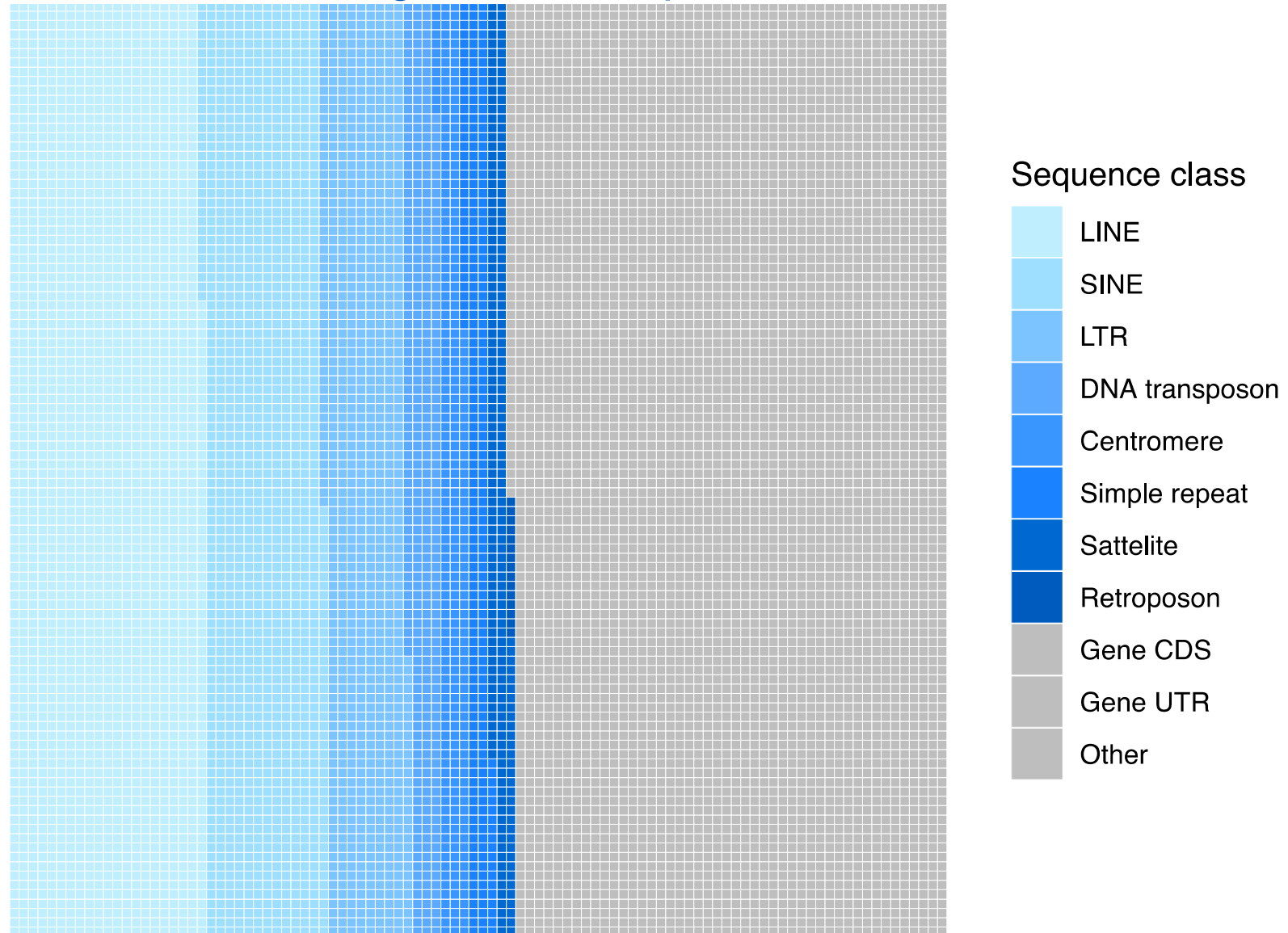




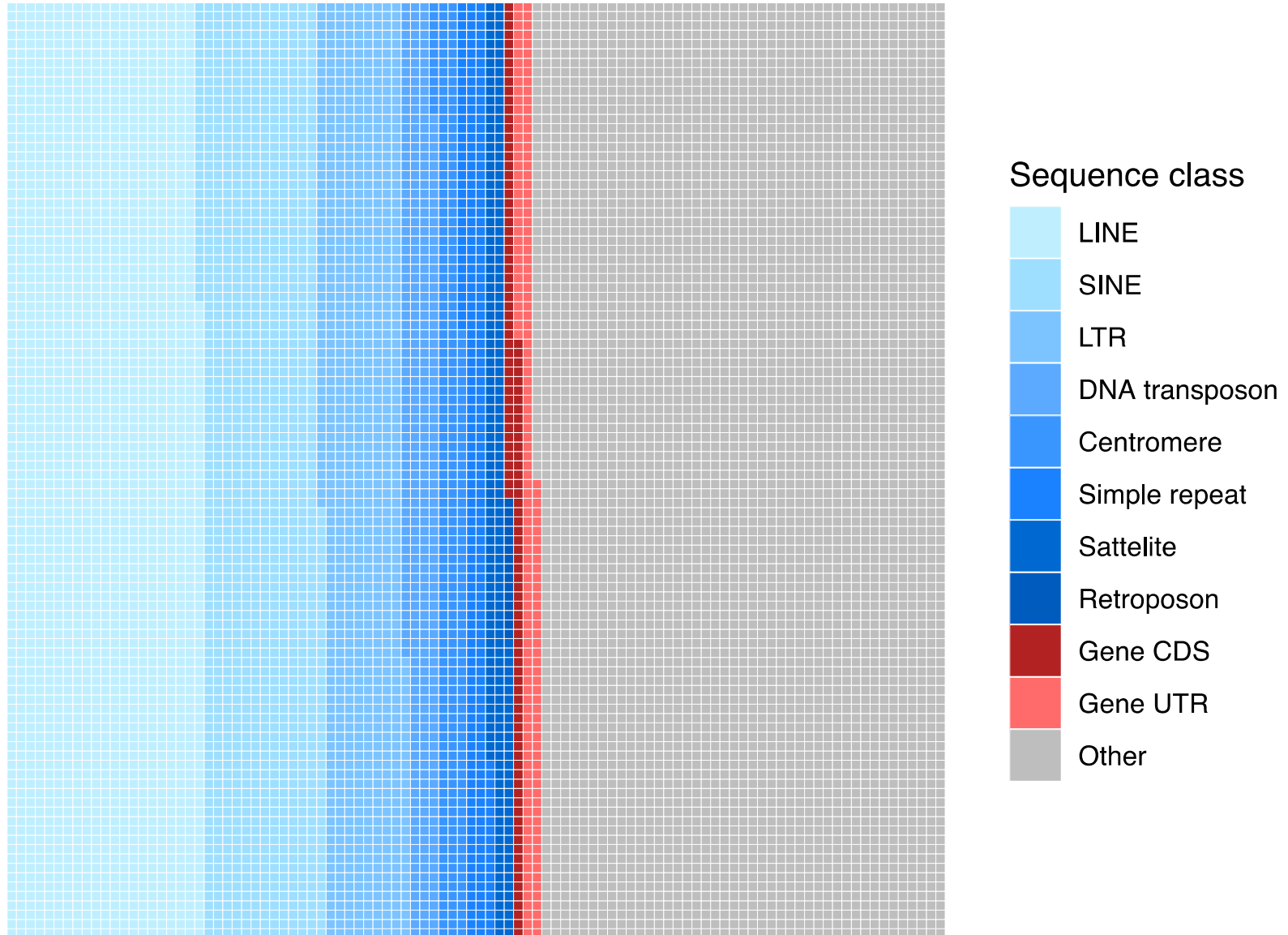
One block is 0.01%
of our genome

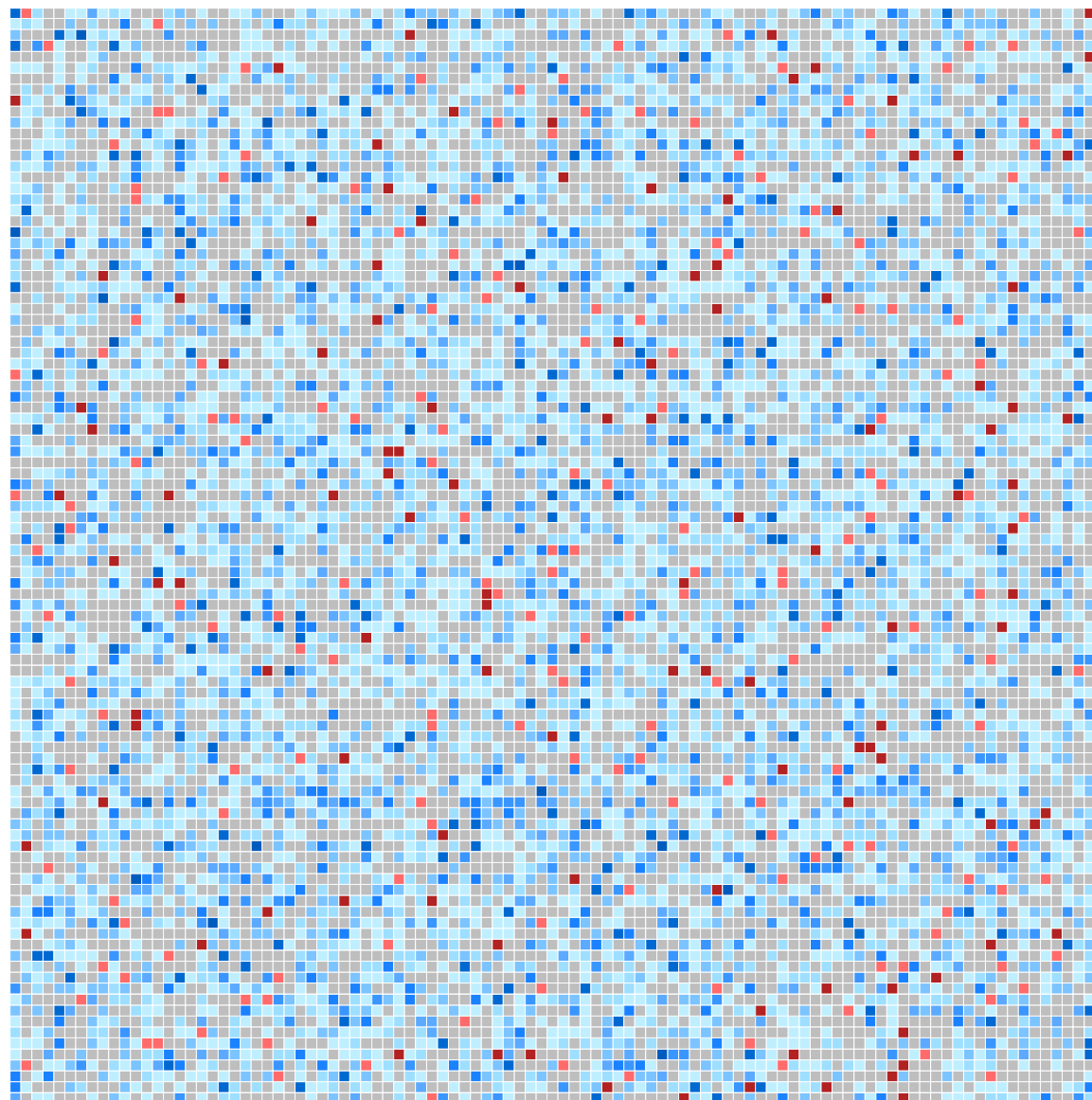


43% of the genome is repetitive!

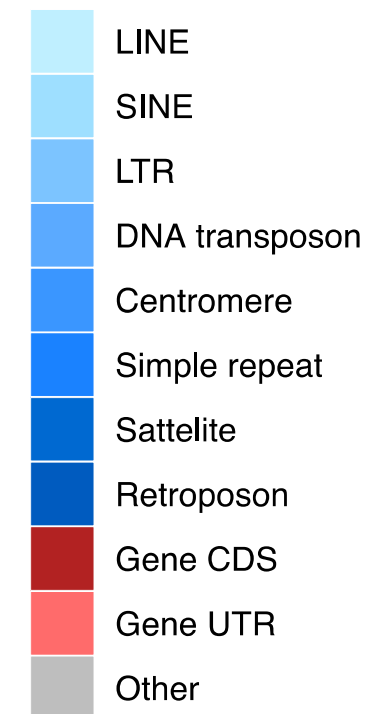


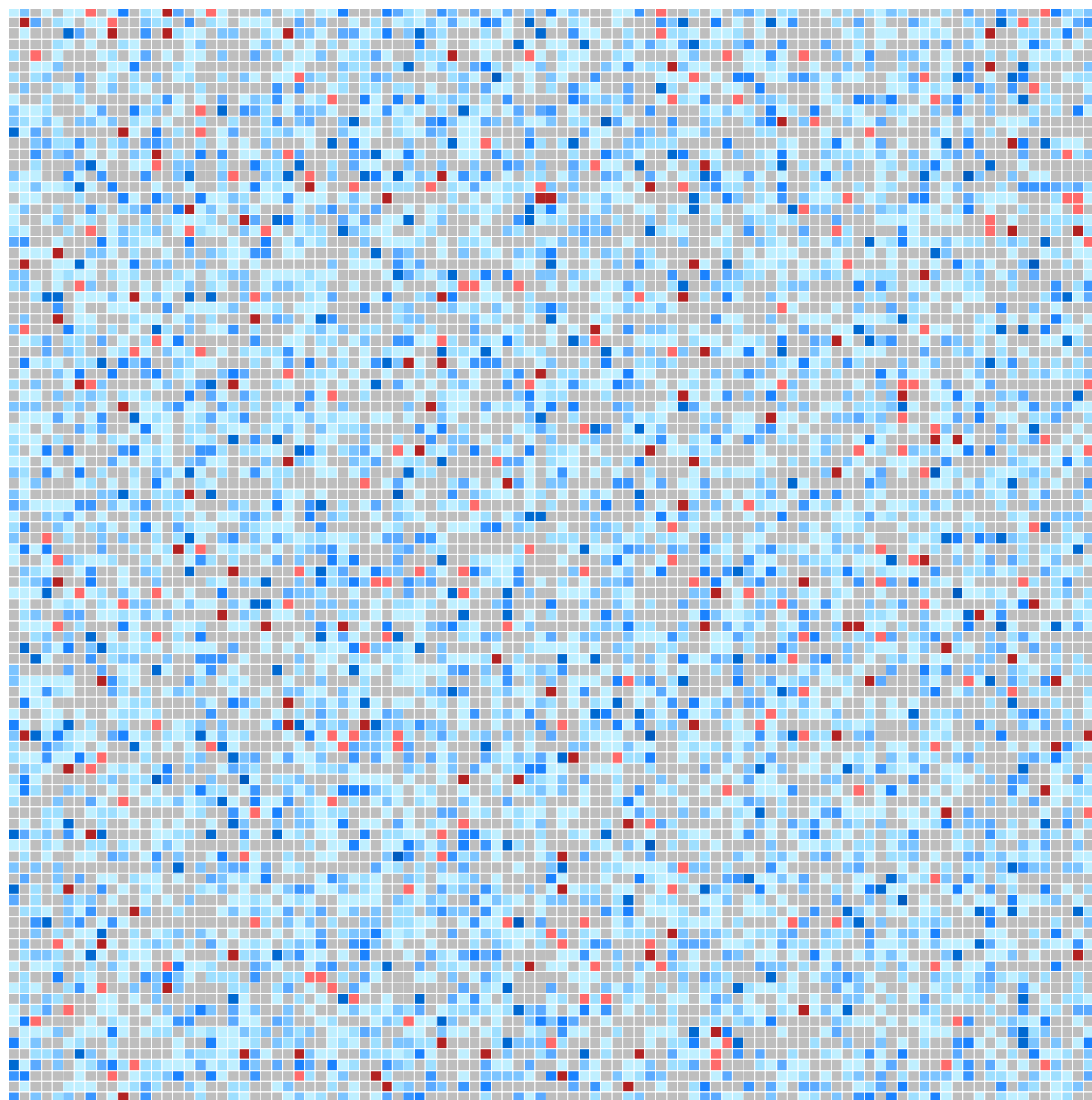
Only 1.2% of the genome is protein coding



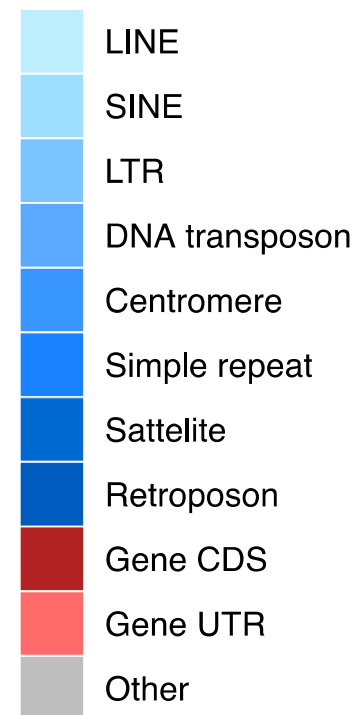


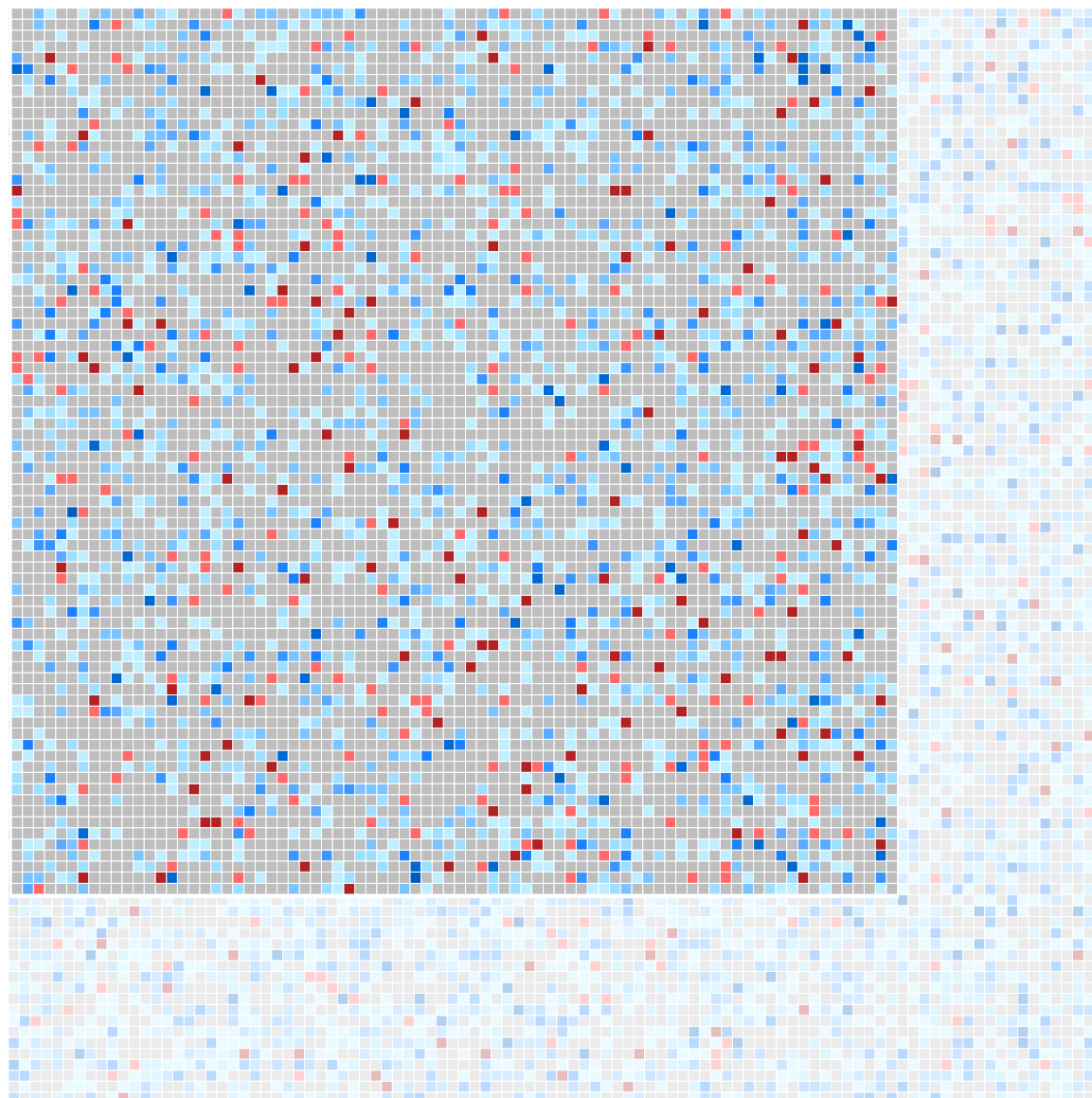
Sequence class



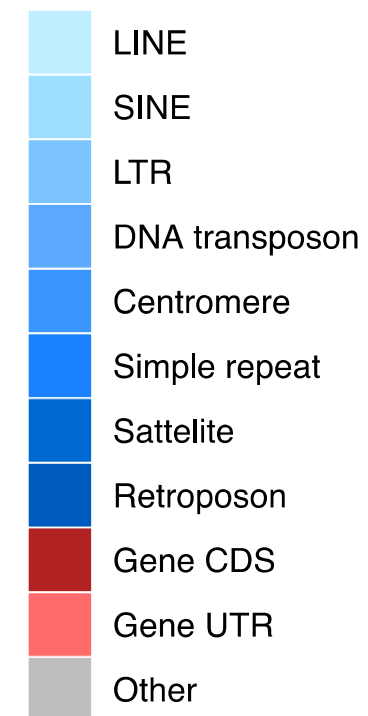


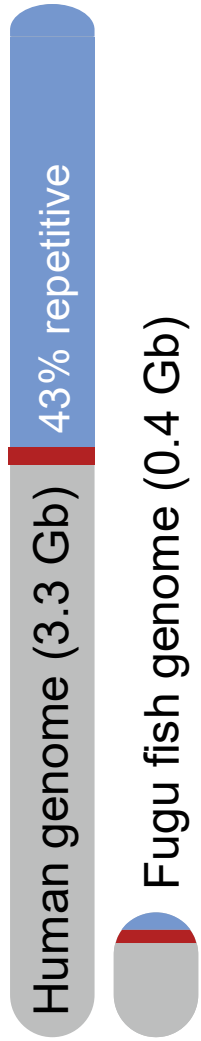
Sequence class



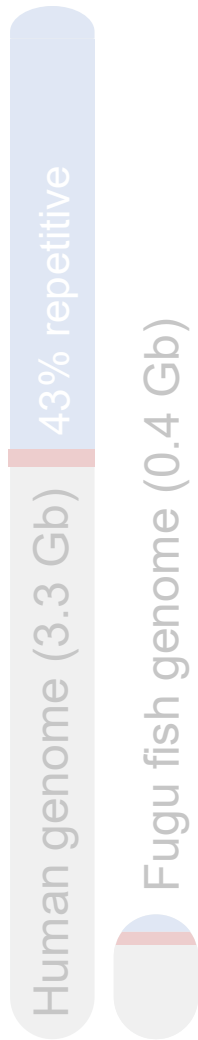


Sequence class



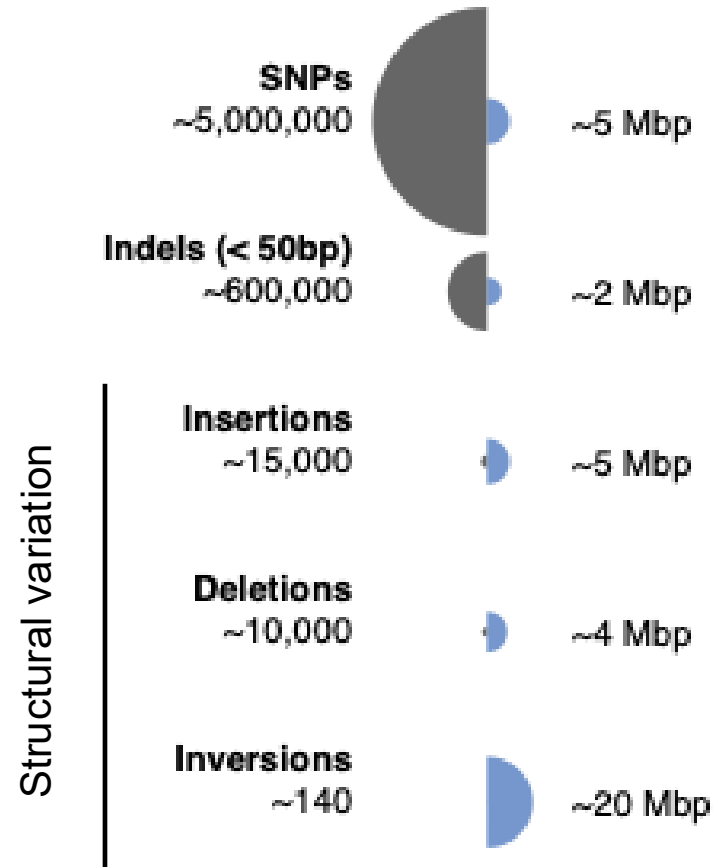


Genome constraints
Minimal genomes



Genome constraints
Minimal genomes

SVs affect more bases in the human genome than any other mutation class



Porubsky & Eichler, 2024

SVs are associated with disease

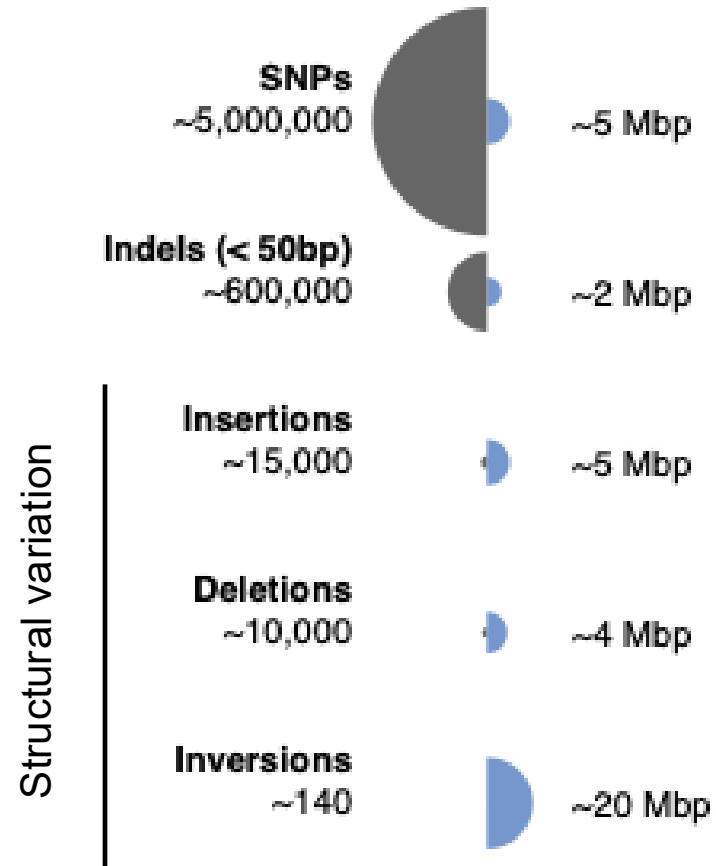
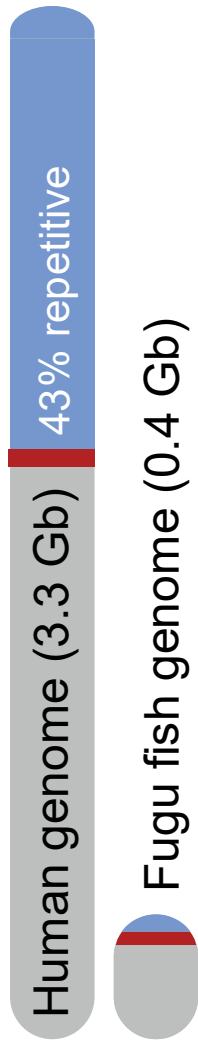
Down Syndrome
Schizophrenia
Hemophilia A
And many more
Weischenfeldt et al., 2013

SVs drive cancers

Copy-number drivers: 3.4
Coding point mutations: 2.6

Aaltonen et al., 2020

Genome constraints Minimal genomes



Porubsky & Eichler, 2024

Clinical interpretation of structural variants



Brix et al., 2024

Training/testing for AI

The one human genome and its natural variation will not be enough to understand genome architecture

What if we could experimentally induce SVs to study structurally distinct human genomes?

Porubsky & Eichler, 2024

Bixi et al., 2024

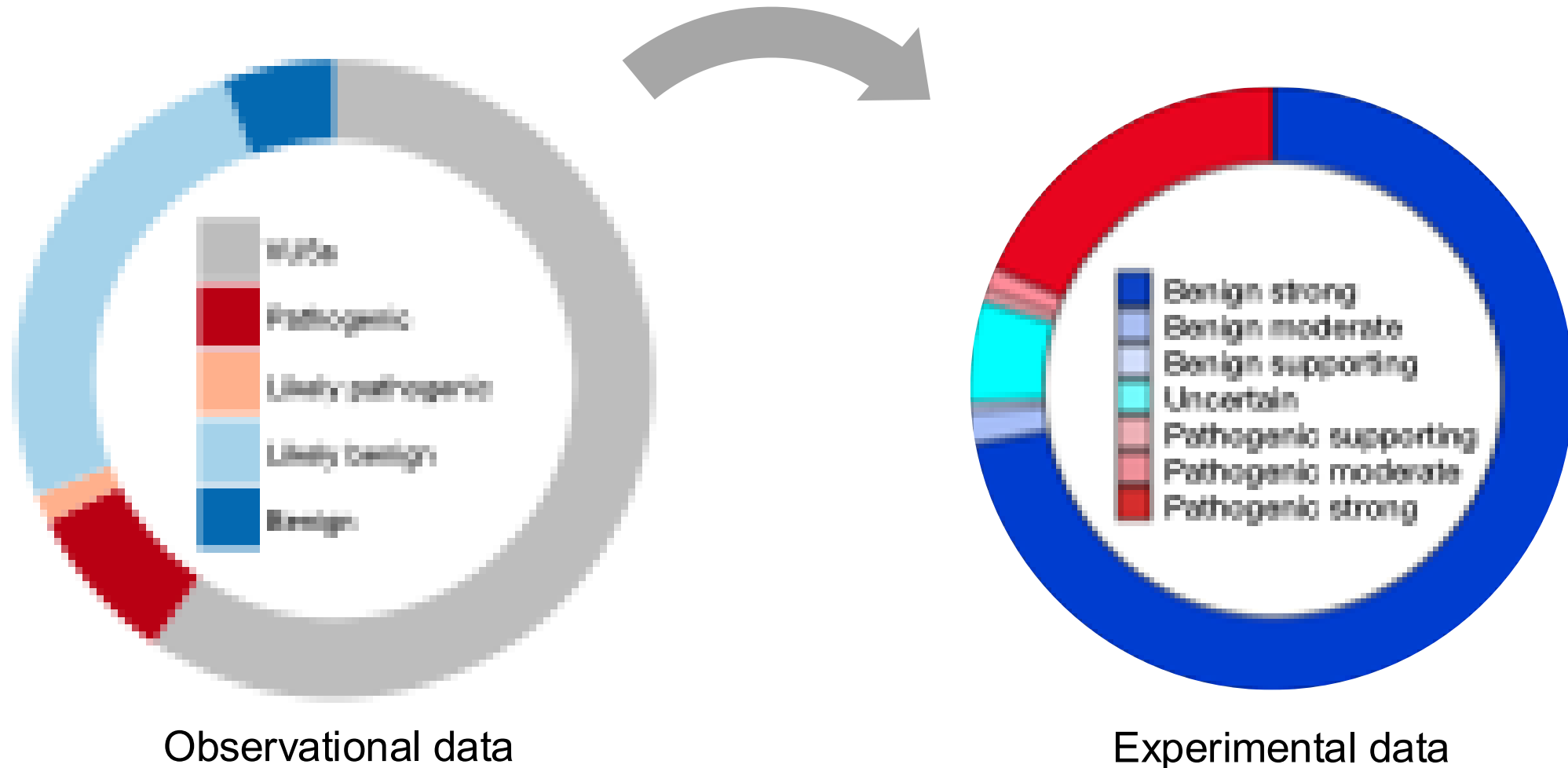
Genome constraints
Minimal genomes

Clinical interpretation
of structural variants

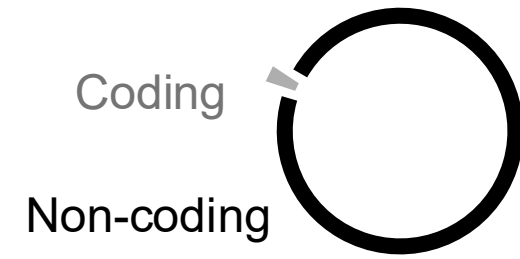
Training/testing
for AI

Experimental perturbation has worked phenomenally at local scale

Example: Saturation genome editing of *BRCA2*



How to scale experimental perturbation to the vast human genome?



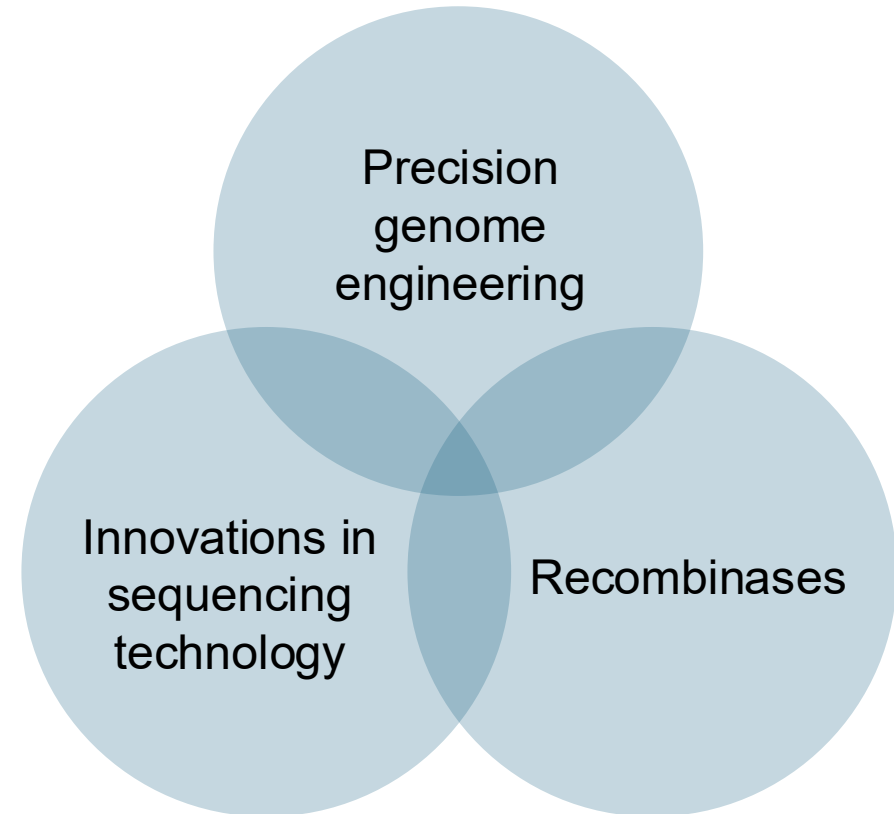
Understanding the genome at a local scale

Saturation genome editing
Multiplexed reporter assays
Knockout screens



Understanding the large scale

Few tools and a limited understanding

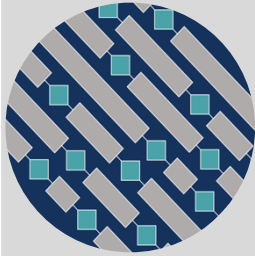




Optimizing genome editors
How do we optimally insert sequences?



Koeppel, Weller,
Peets et al., 2023



Engineering regulatory landscapes
How do enhancer clusters work?



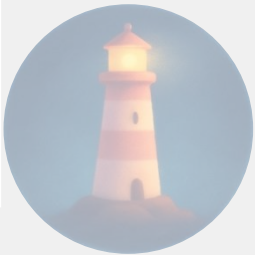
Koeppel, Murat,
et al., 2025



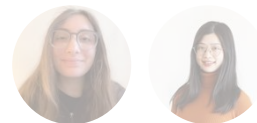
Scrambling entire human genomes
What is the fitness landscape of SVs?



Koeppel, Ferreira,
et al., 2025



High-resolution deletion scanning
How much of the genome is essential?



In preparation

Research program: Engineering SV to understand genome function

Prime editors + recombination = structural diversity



Prime editors + recombination = structural diversity



add Cre recombinase

Inversion



Deletion



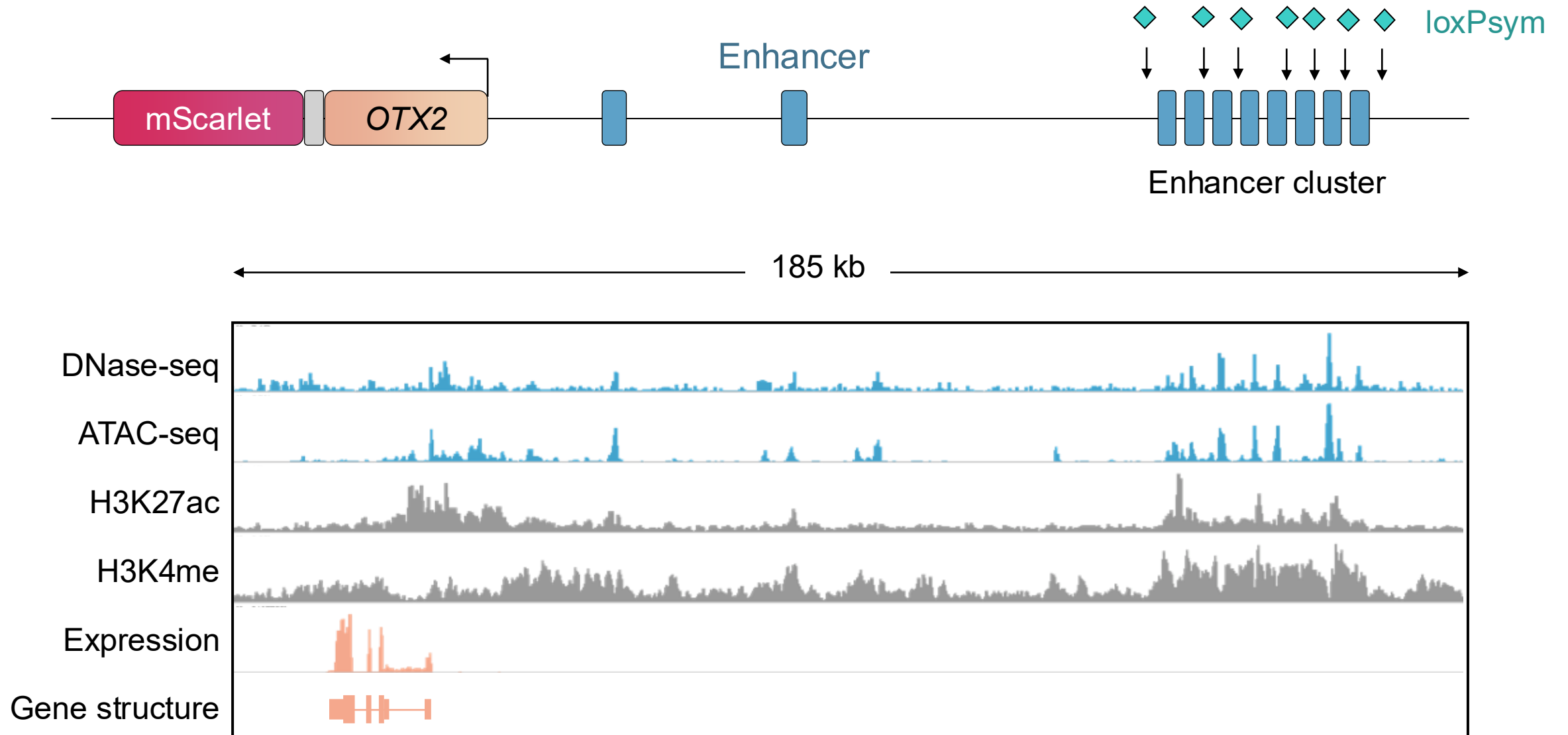
Complex



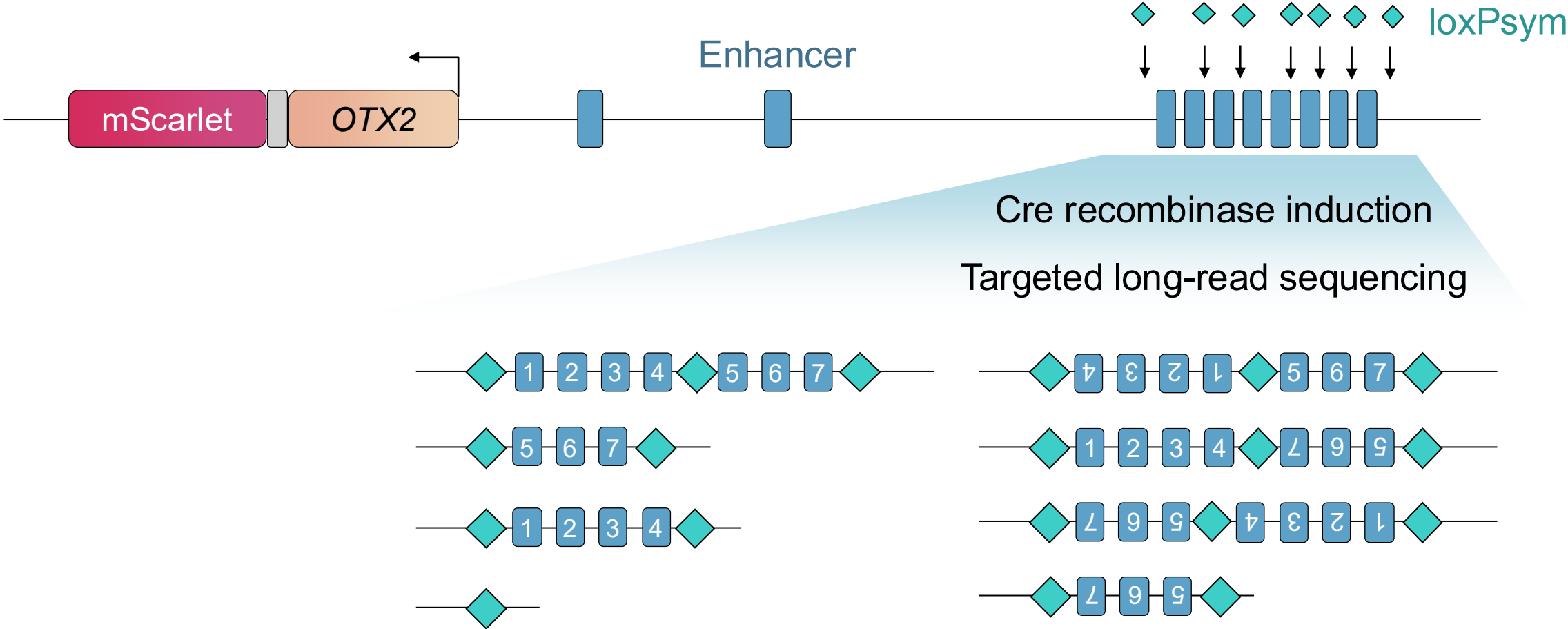
Complex



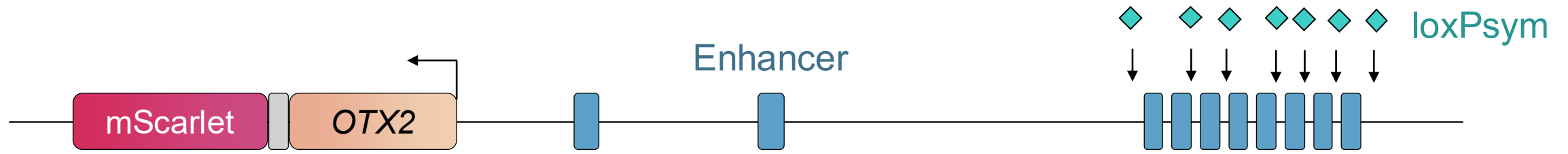
Targeted randomization allows context and scale



Randomizing an enhancer cluster

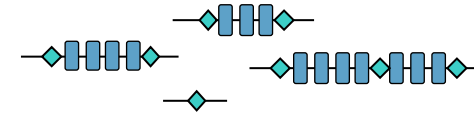


Randomizing an enhancer cluster



Day 0

Induce with 2 μ M Cre for 4h



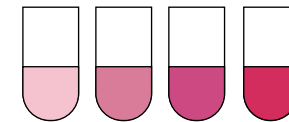
Day 14

Sort cells based on expression



Day 20

Harvest cells & extract DNA

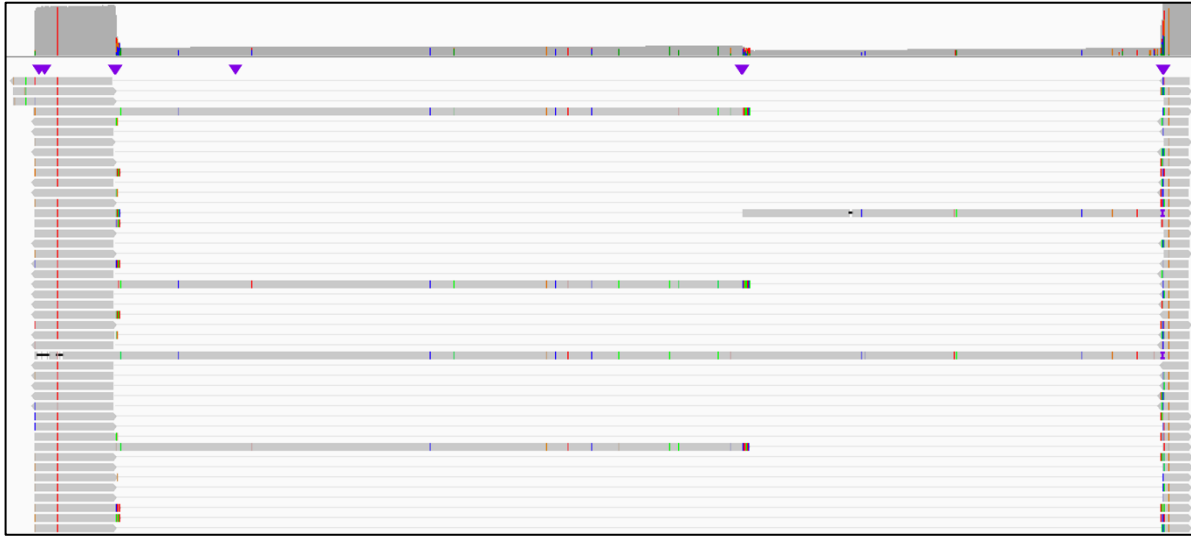


Sequence enhancer architecture

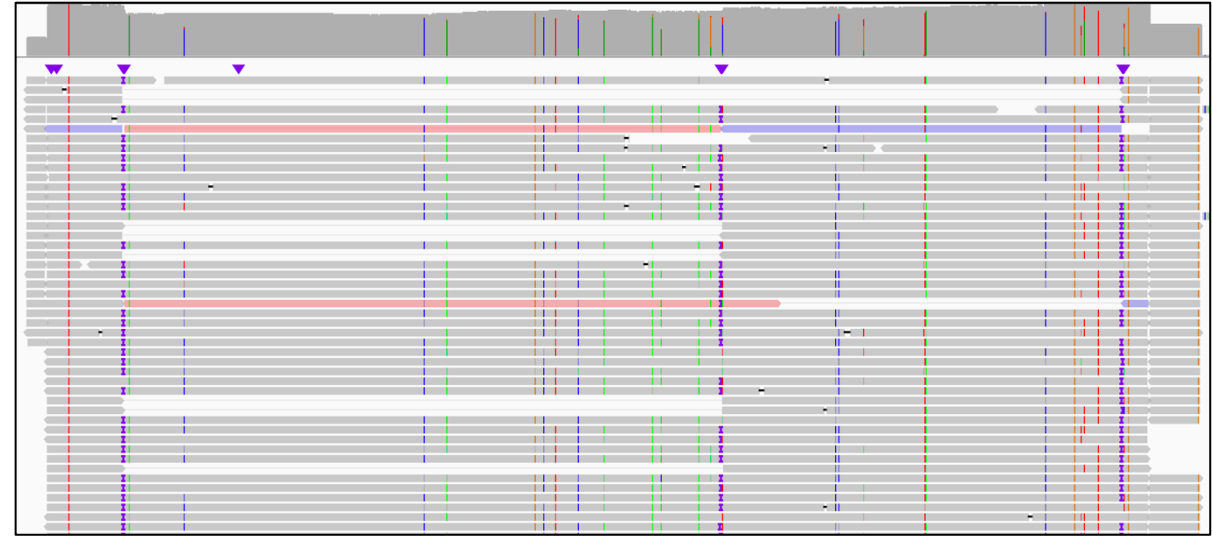


Structural variants differ between sorting gates

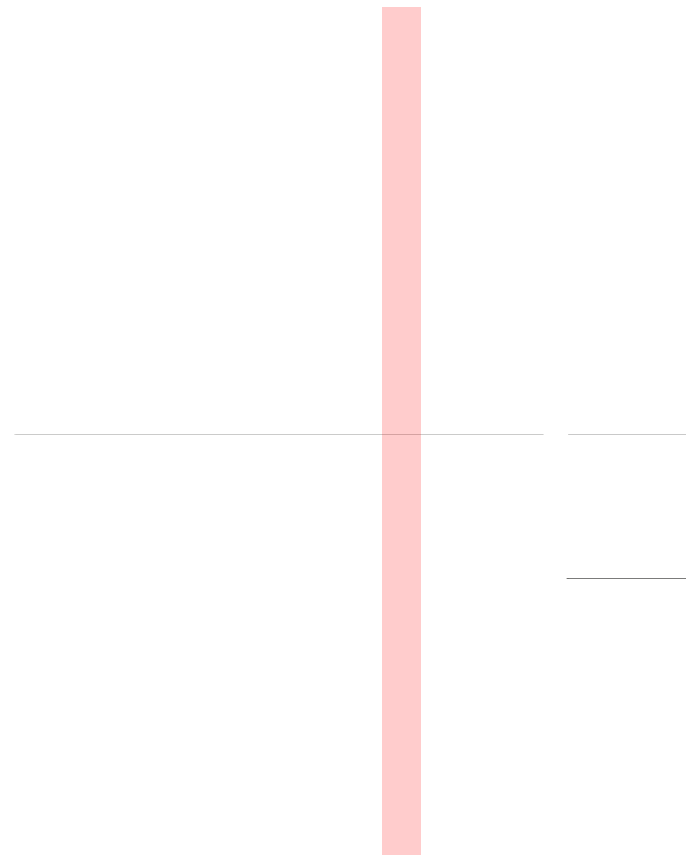
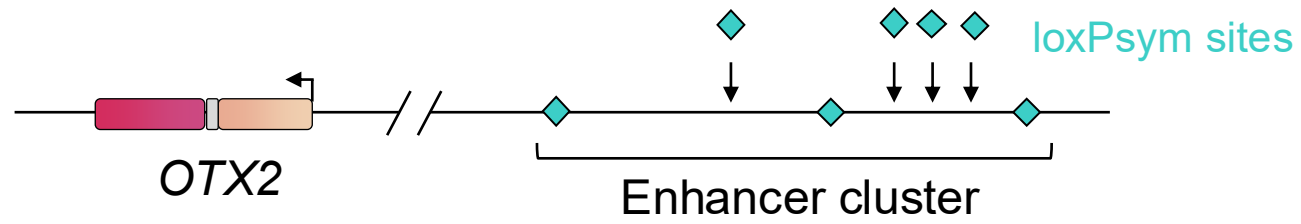
Dim gate ($\downarrow OTX2$)



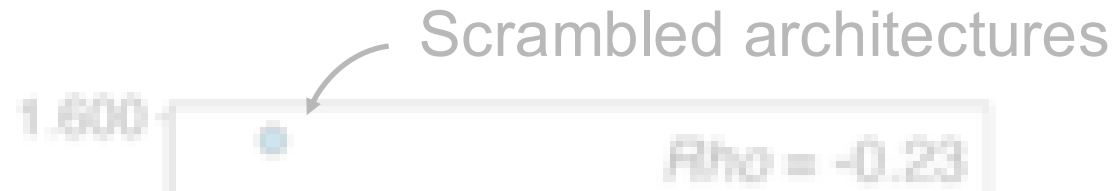
Bright gate ($\uparrow OTX2$)



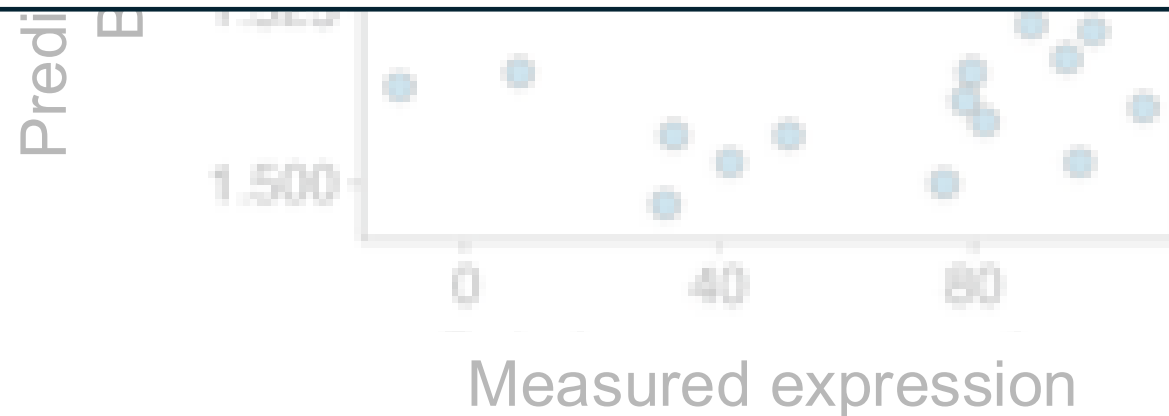
A single region drives the majority of *OTX2* expression



Genomic AI models cannot predict the impact of SVs



We need to measure the expression impacts of many more SVs and train better models!

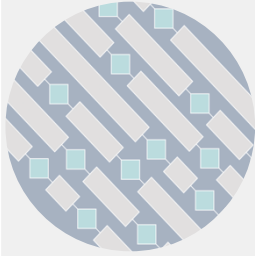




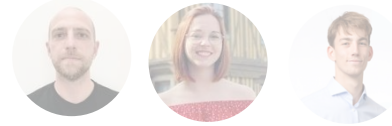
Optimizing genome editors
How do we optimally insert sequences?



Koeppel, Weller,
Peets et al., 2023



Engineering regulatory landscapes
How do enhancer clusters work?



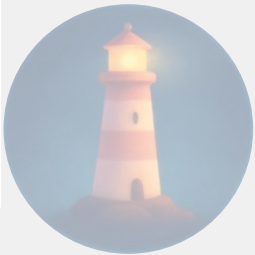
Koeppel, Murat,
et al., 2025



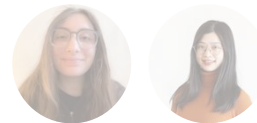
Scrambling entire human genomes
What is the fitness landscape of SVs?



Koeppel, Ferreira,
et al., 2025



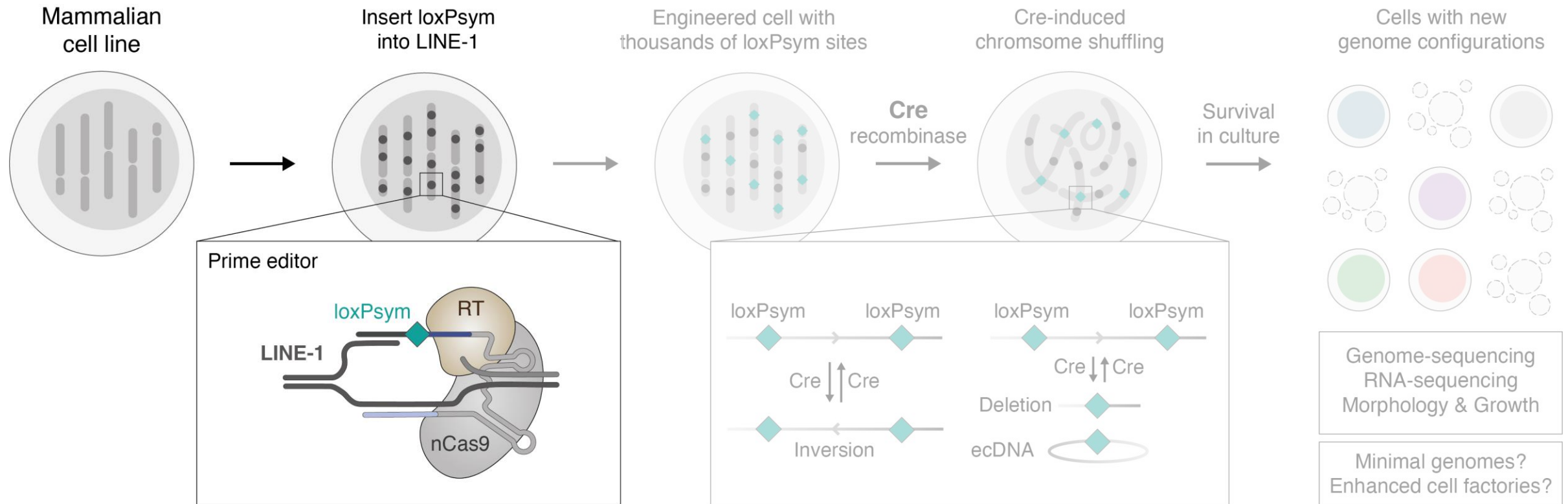
High-resolution deletion scanning
How much of the genome is essential?



In preparation

Research program: Engineering SV to understand genome function

Combining prime editing with the targeting of repetitive elements



Simultaneously targeting thousands of sites across the genome

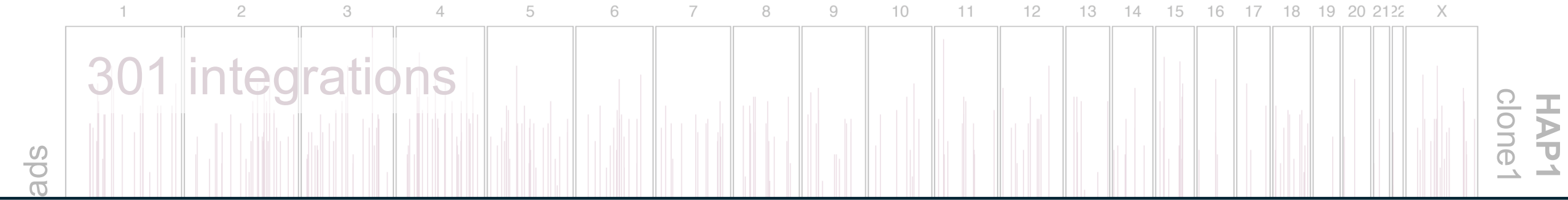
LINE1 retrotransposon



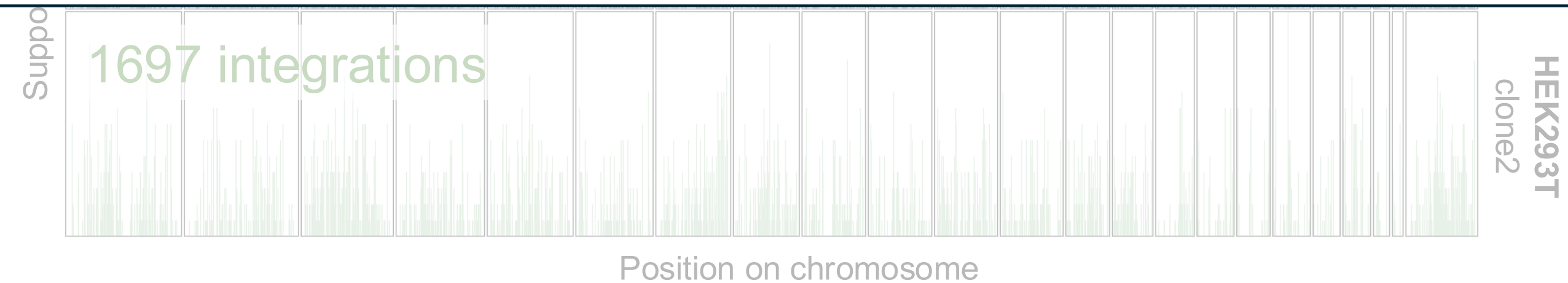
pegRNA protospacer targets

Genomic copies	5832
With 1 mismatch	11090
With 2 mismatches	14297

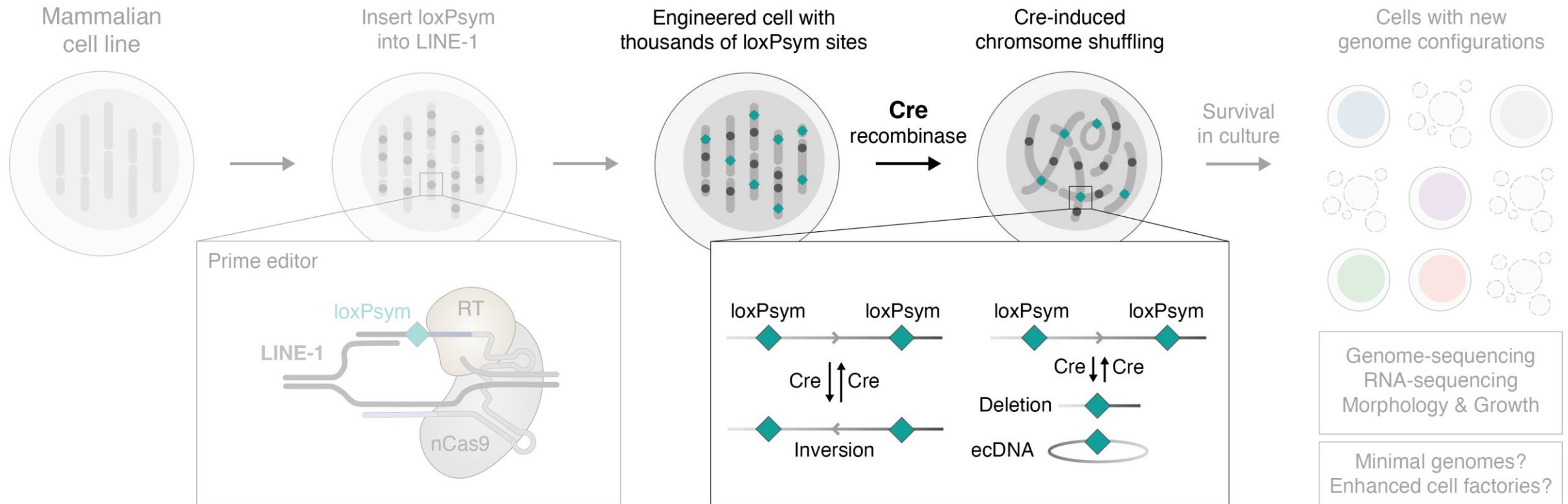
We can integrate hundreds of loxPsym sites into a single genome



A lot of optimization later!

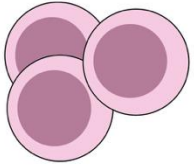


Combining prime editing with the targeting of repetitive elements

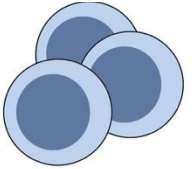


Cre recombinase treatment induced thousands of rearrangements

HAP1 301 loxPsym



HEK293T 638 loxPsym



Cre protein
4 hours

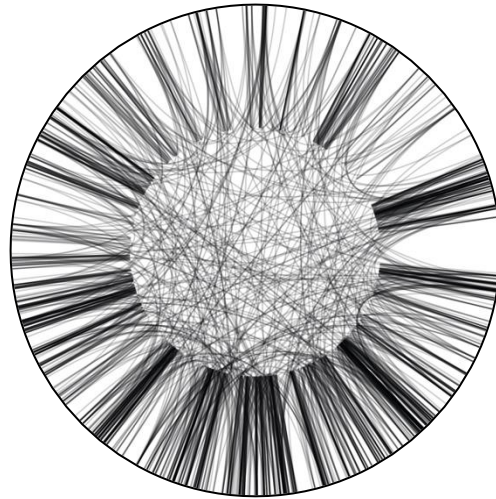


Growth
10 hours

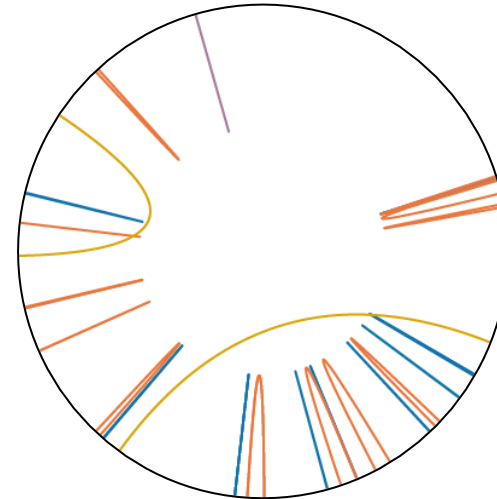
Long-read
sequencing

Experimental SVs can uncover the “missing” SV space

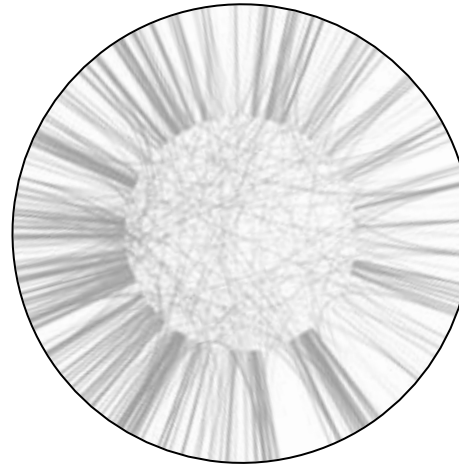
All SVs that arise
hidden, pre-selection space



SVs we observe by sequencing
post-selection survivors



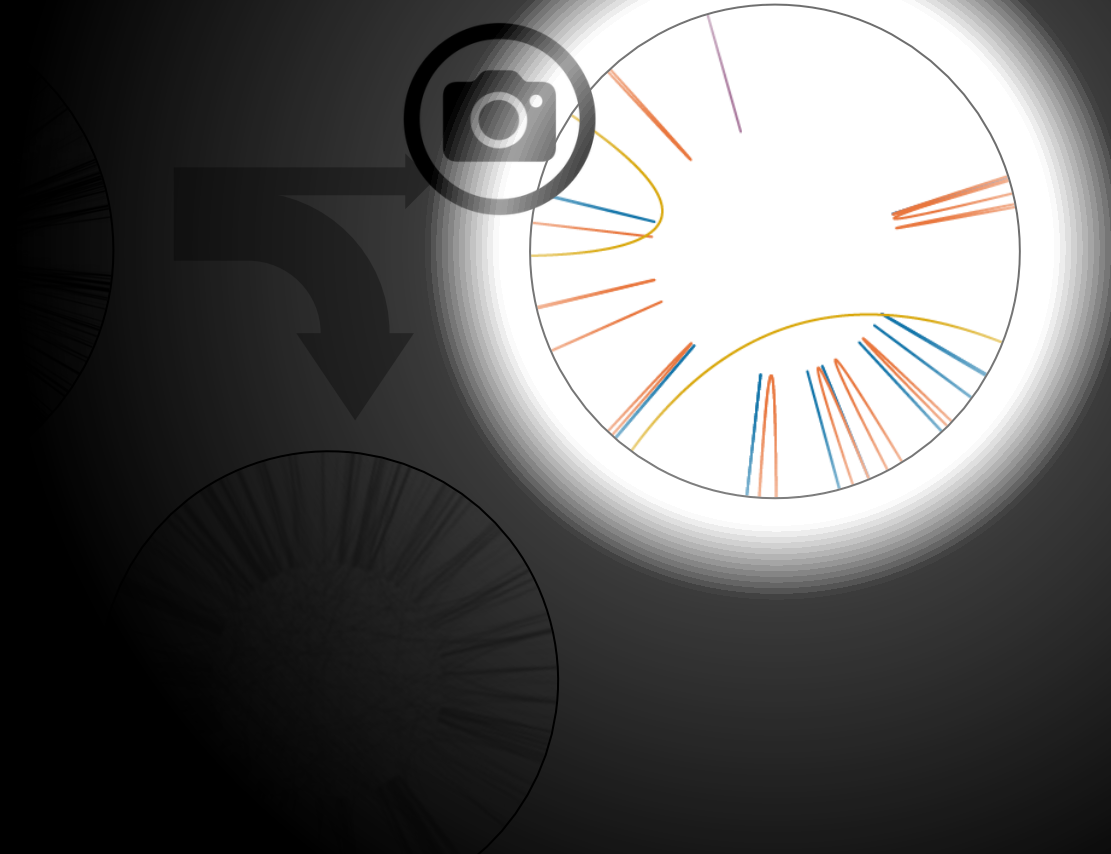
SVs lost to
negative selection



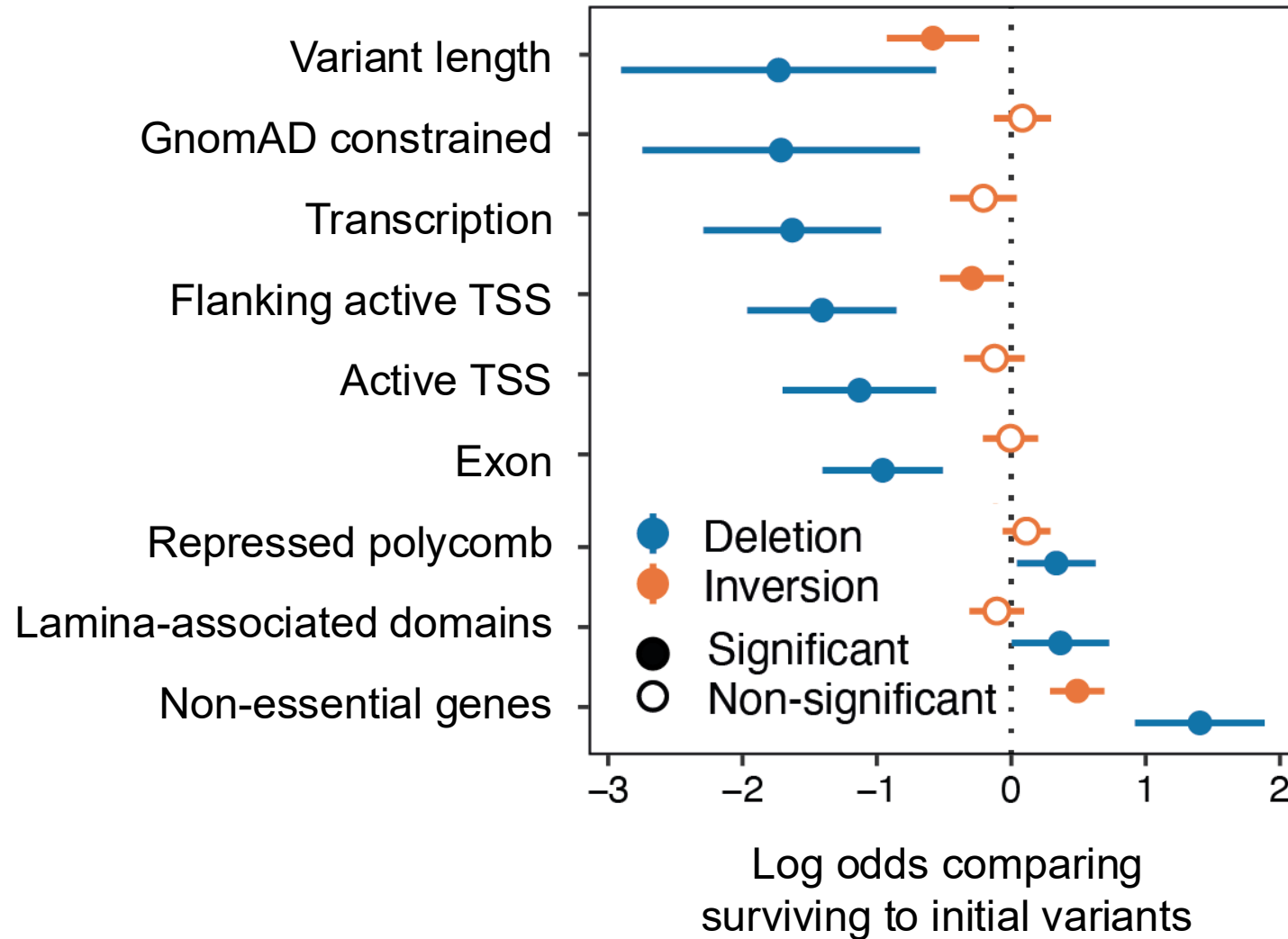
the “missing” SV space

SV space

**SVs we observe by sequencing
post-selection survivors**

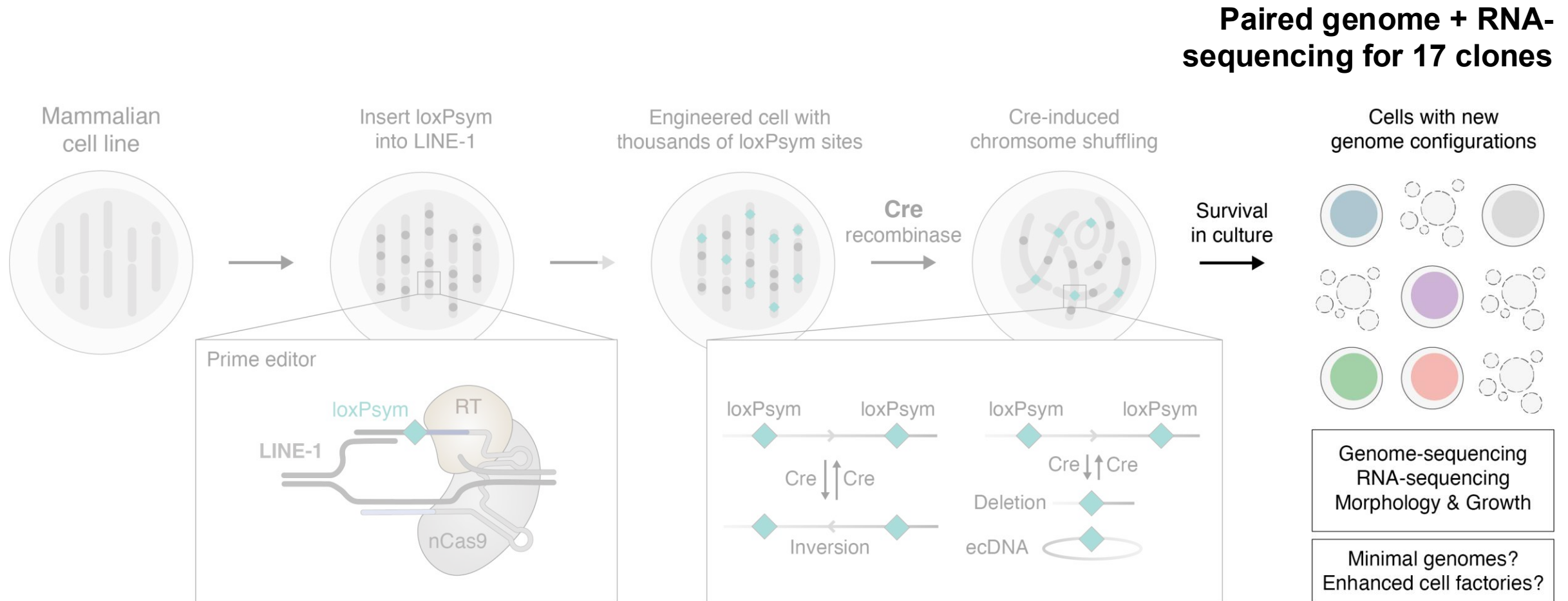


Selection shapes surviving variants



Comparing 159 **surviving** deletions/inversions to 651 initial variants

Combining prime editing with the targeting of repetitive elements



Surviving clones with megabase sized variants

HAP1 scrambled clone 15

Chr	Start [Mb]	Variant type	Variant size [kb]	Genes affected
Chr2	18.0	Deletion	3197	10
ChrX	70.6	Deletion	241	0

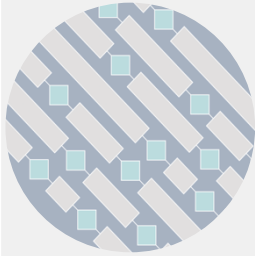


Optimizing genome editors

How do we optimally insert sequences?

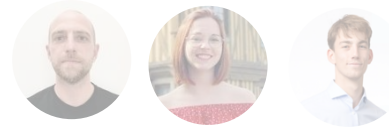


Koeppel, Weller,
Peets et al., 2023



Engineering regulatory landscapes

How do enhancer clusters work?



Koeppel, Murat,
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Scrambling entire human genomes

What is the fitness landscape of SVs?

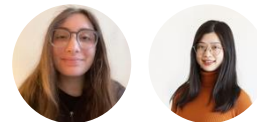


Koeppel, Ferreira,
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High-resolution deletion scanning

How much of the genome is essential?



In preparation

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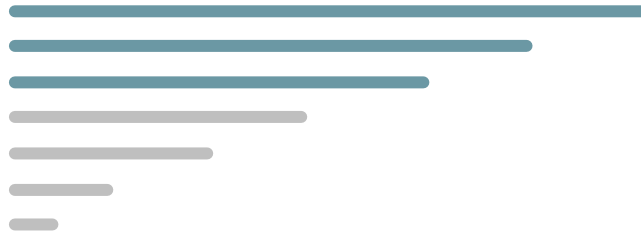
How to overcome limitations of previous shuffle/scramble attempts?

Scramble/shuffle approach

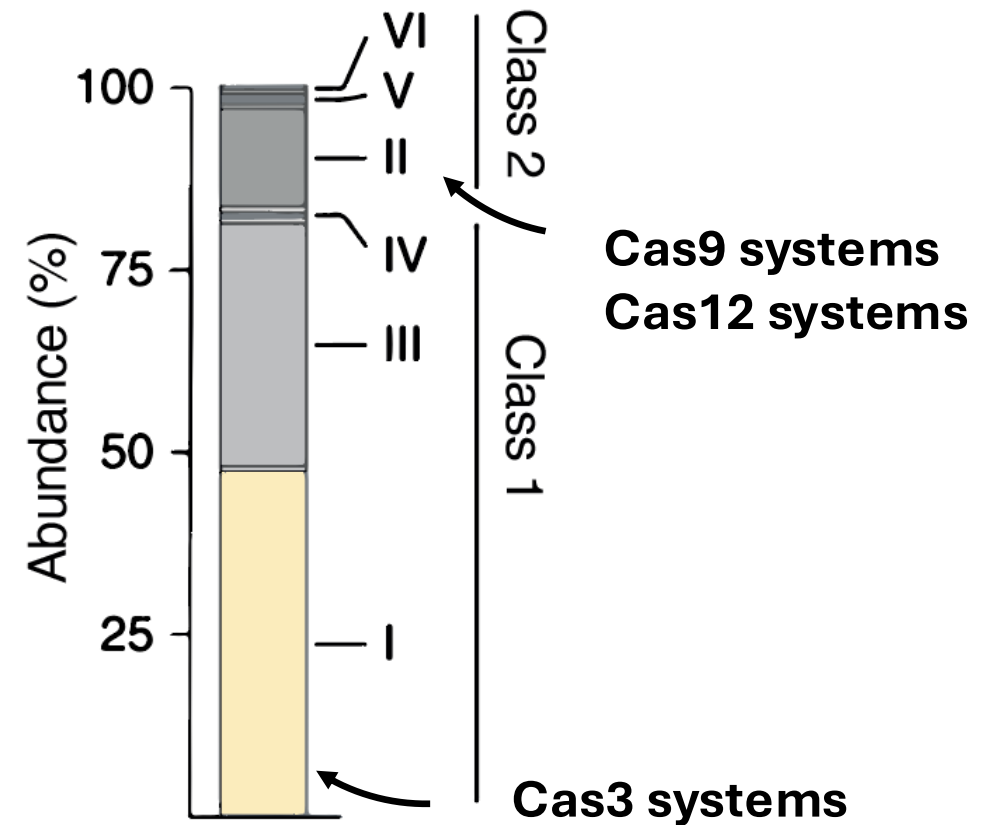
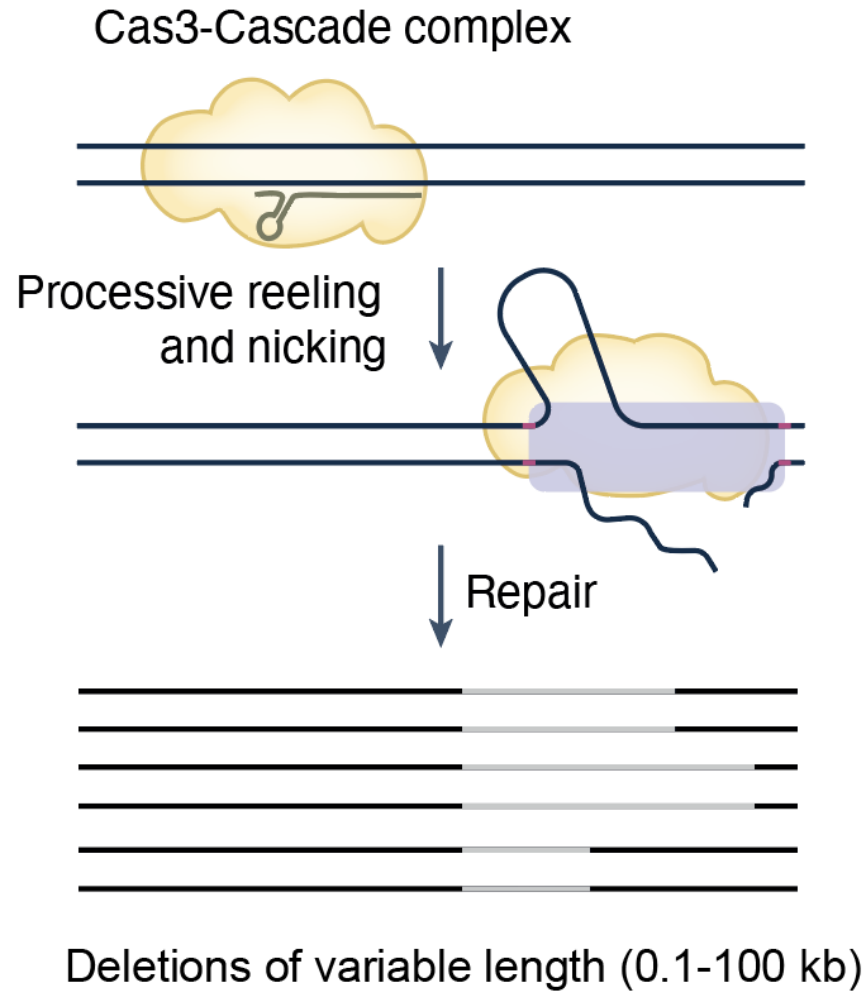


Important sequence element

Deletion scanning
approach



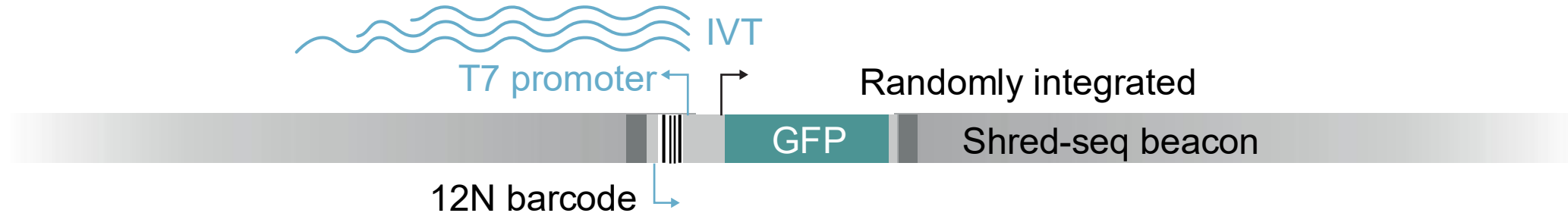
Cas3 initiates large and unidirectional deletions



Adapted from Cameron et al., 2019

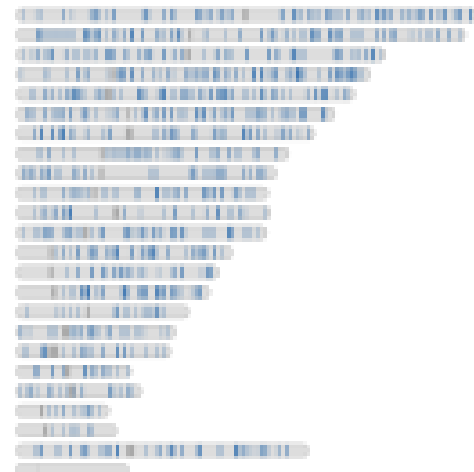
Deletions are launched from a genomically-integrated beacon

Map Barcodes to genomic locations



Bottleneck cells with integration

4000 bottleneck

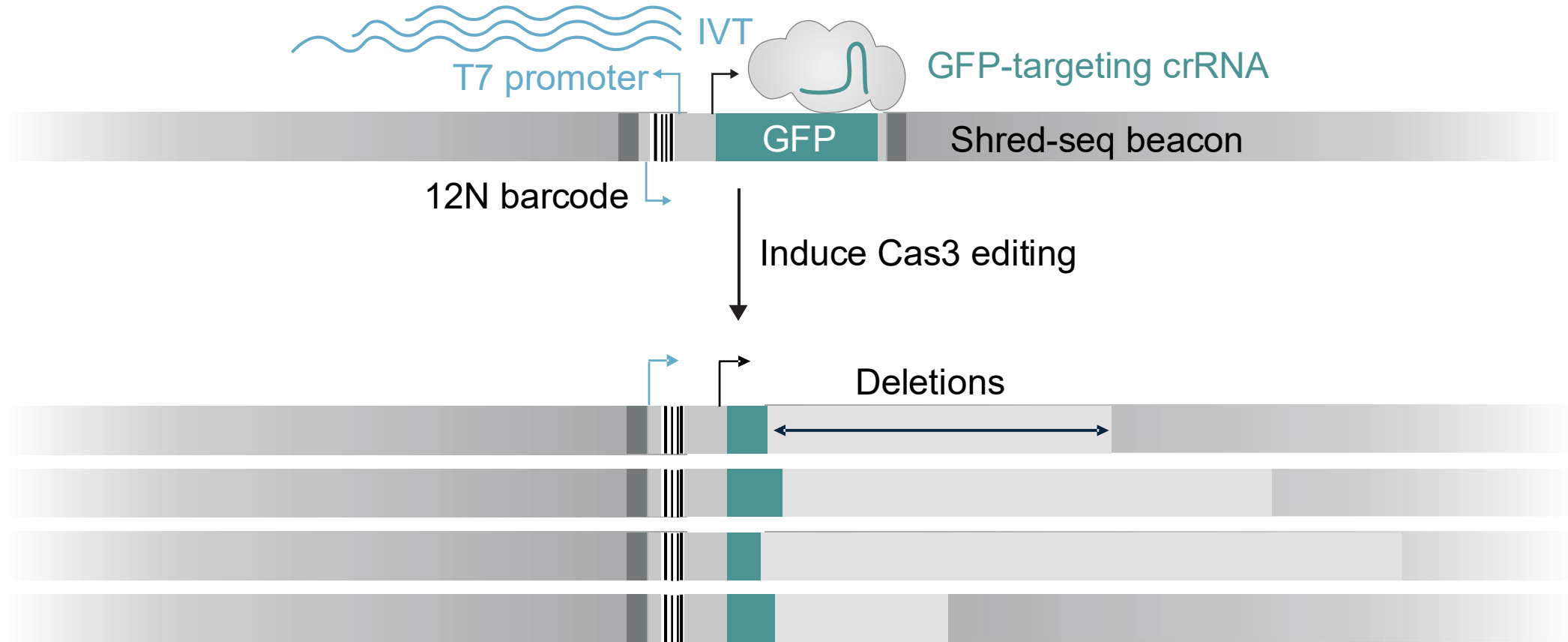


50000 bottleneck



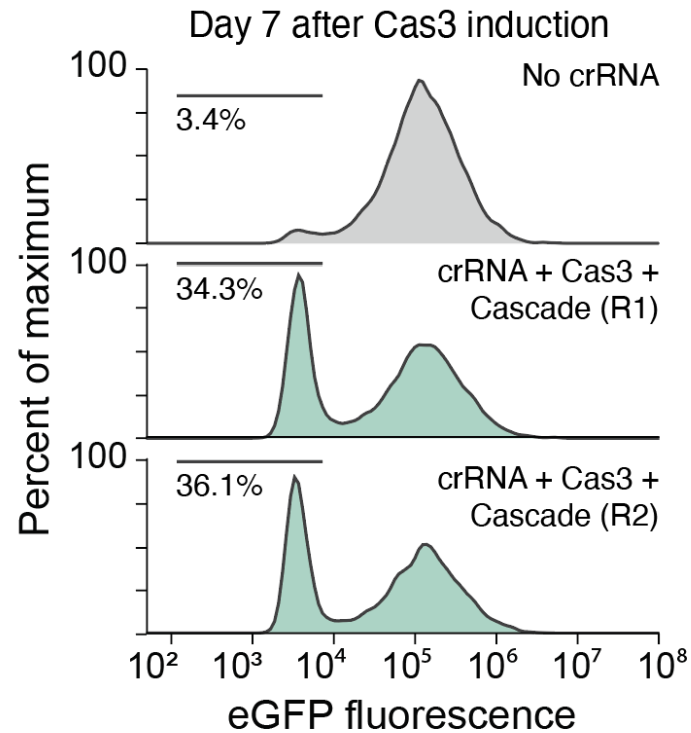
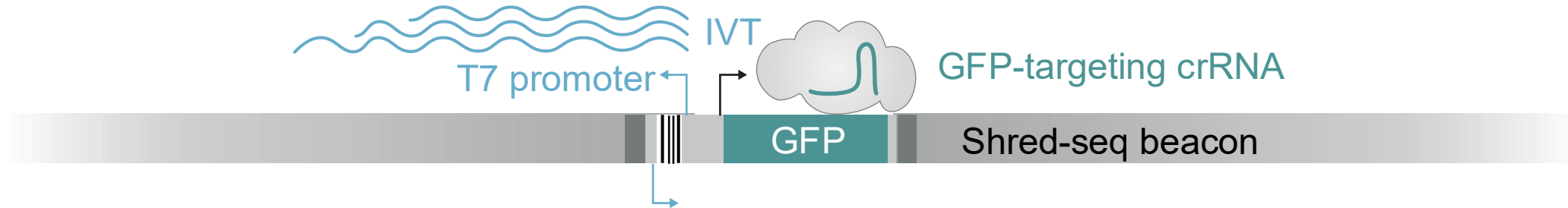
Deletion scanning with Cas3

Map Barcodes to genomic locations



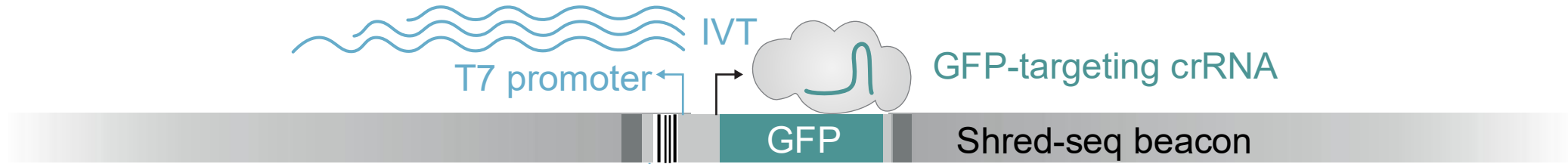
Induction with Cas3 results in 30-40% GFP loss

Map Barcodes to genomic locations



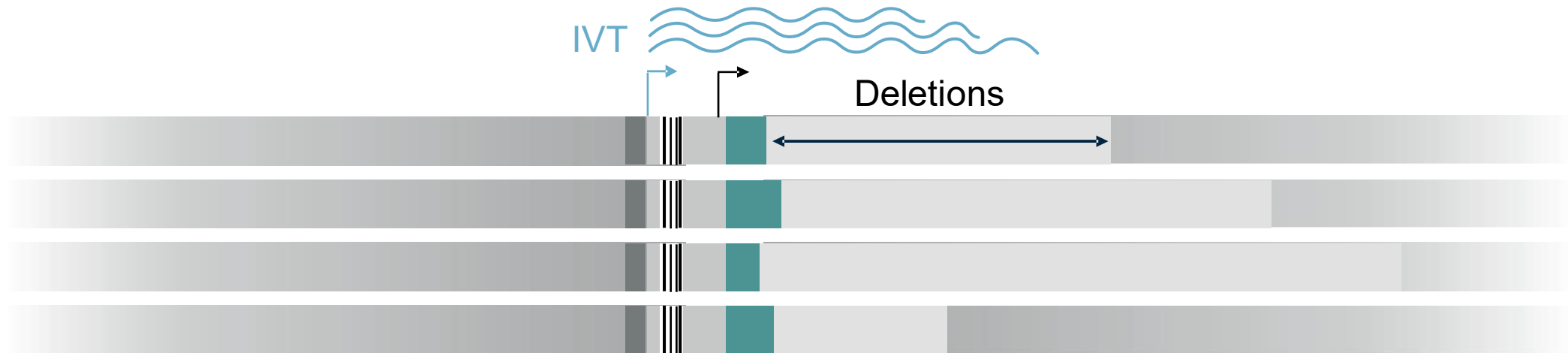
High-throughput deletion mapping

Map Barcodes to genomic locations



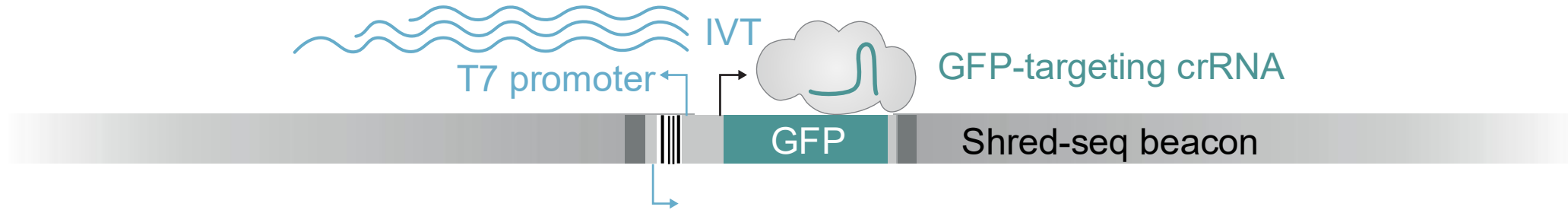
Induce Cas3 editing
Optional: Sort for GFP-loss

Map deletion boundaries

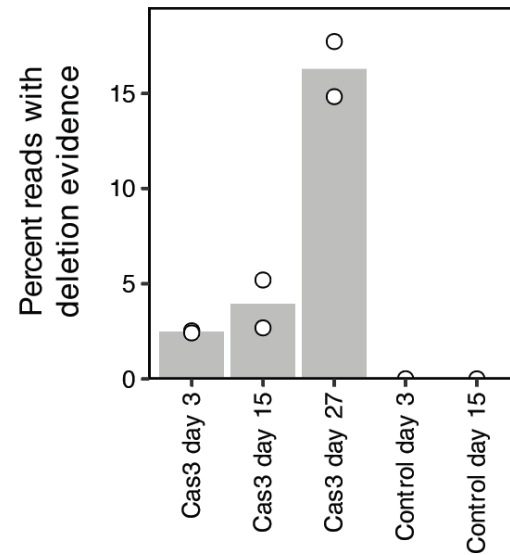


Deletion scanning with Cas3

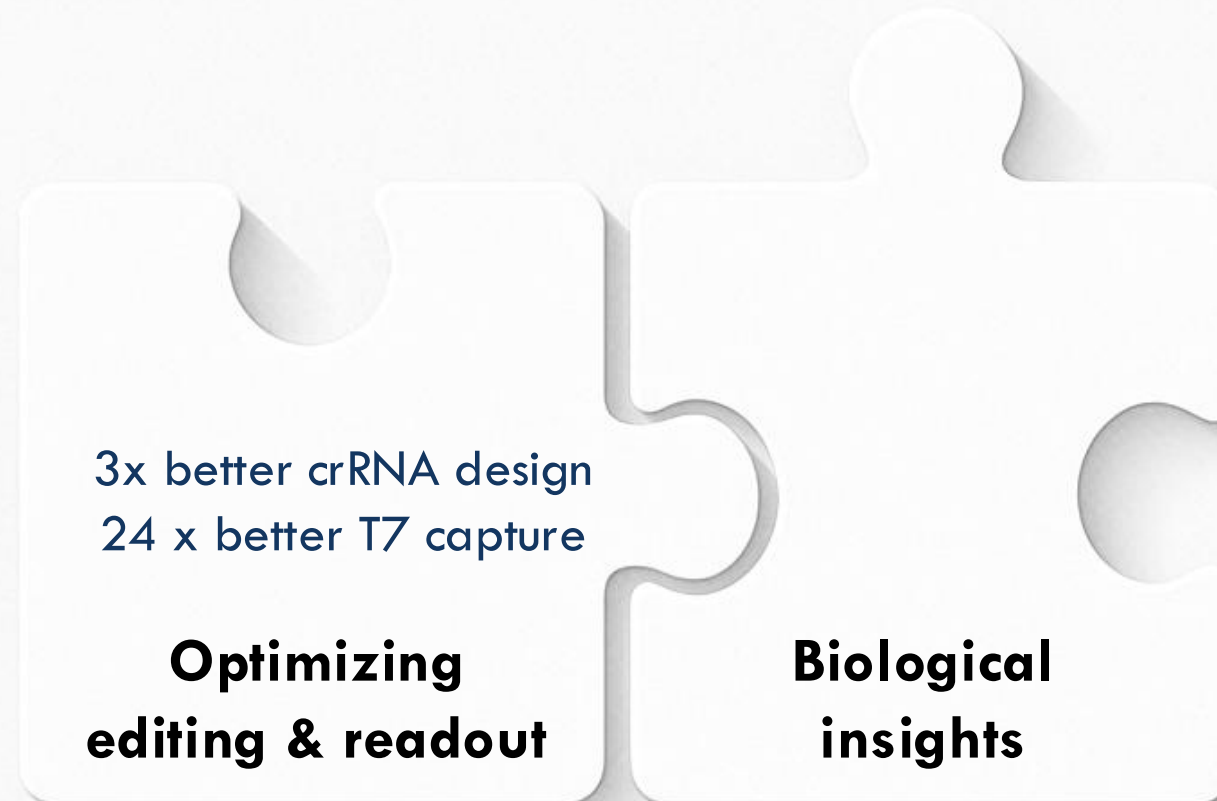
Map Barcodes to genomic locations



Fraction of deletions



Optimizing genome editors and biological insights are pieces of the same puzzle

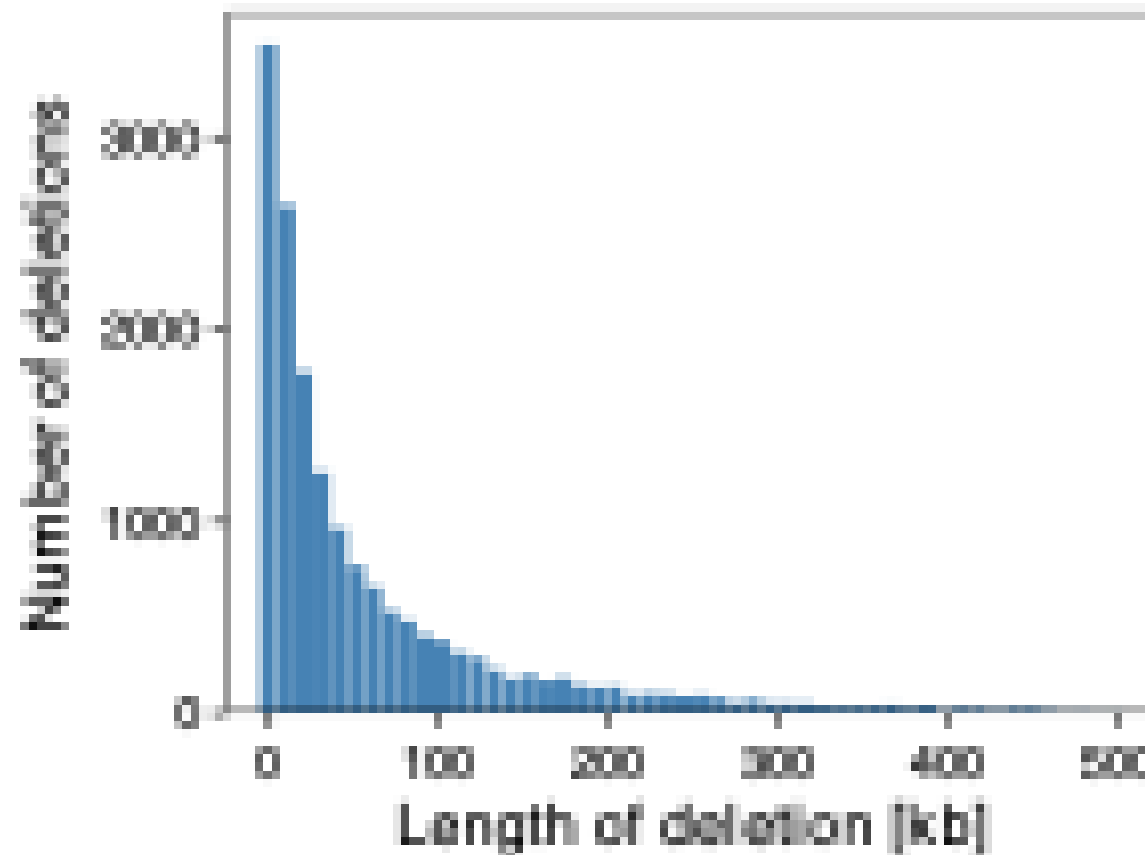


We deleted > 1 Gb of DNA across a pool of cells

Median deletion size: **29 kb**

Total number of unique deletions: **16,882**

Total bases deleted: **1,122,492,294 bp**



Features of surviving SVs

Deletion no.	Day observed	Ploidy of cell	No. essential genes	Predicted deleteriousness	...
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Method

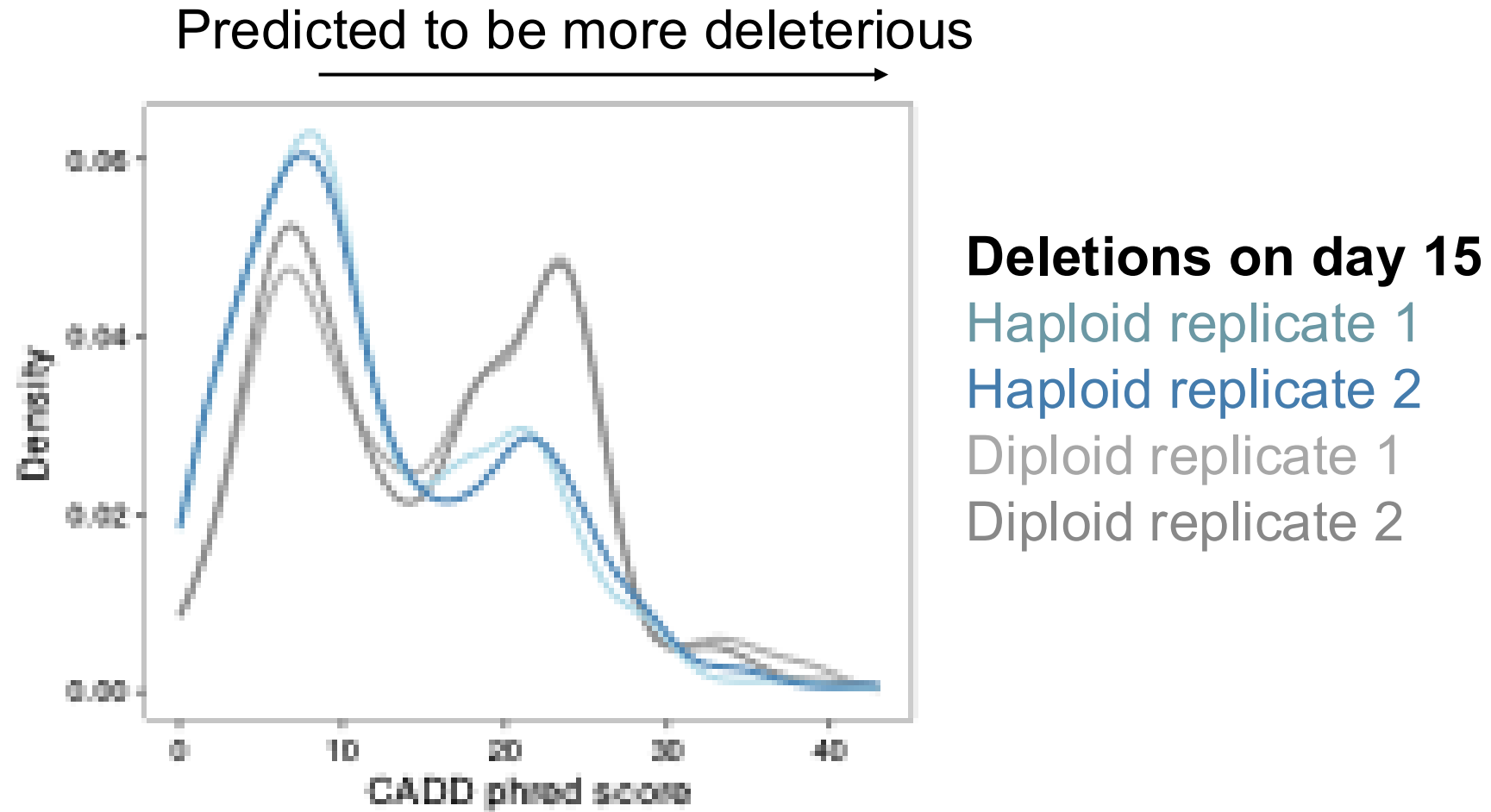
A framework to score the effects of structural variants in health and disease

Philip Kleinert¹ and Martin Kircher^{1,2}

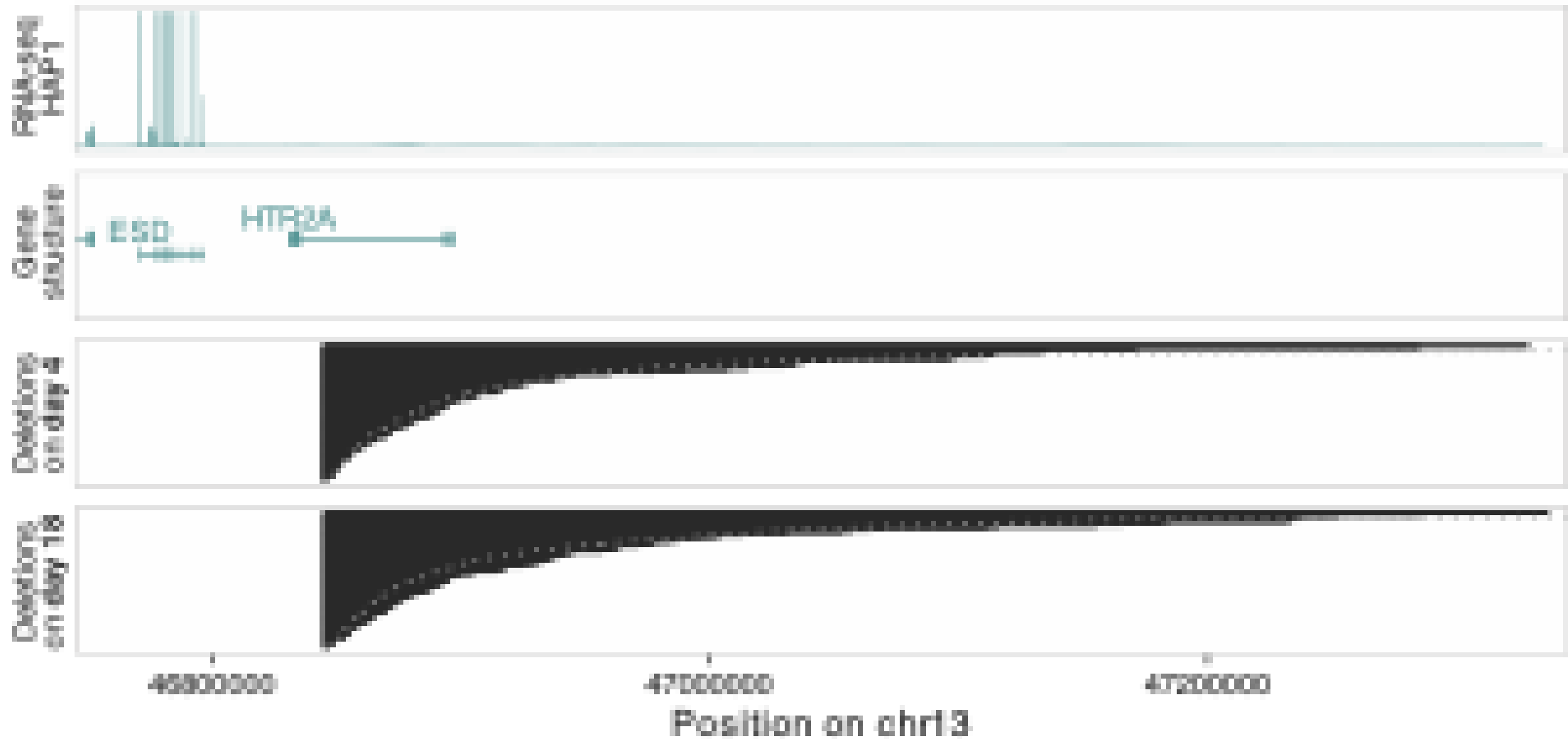
¹Berlin Institute of Health (BIH) at Charité–Universitätsmedizin Berlin, 10117 Berlin, Germany; ²Institute of Human Genetics,
University Medical Center Schleswig-Holstein, University of Lübeck, 23562 Lübeck, Germany

7	3	haploid	2	32	
...	15	diploid	1	20	
16,882	3	haploid	0	3	

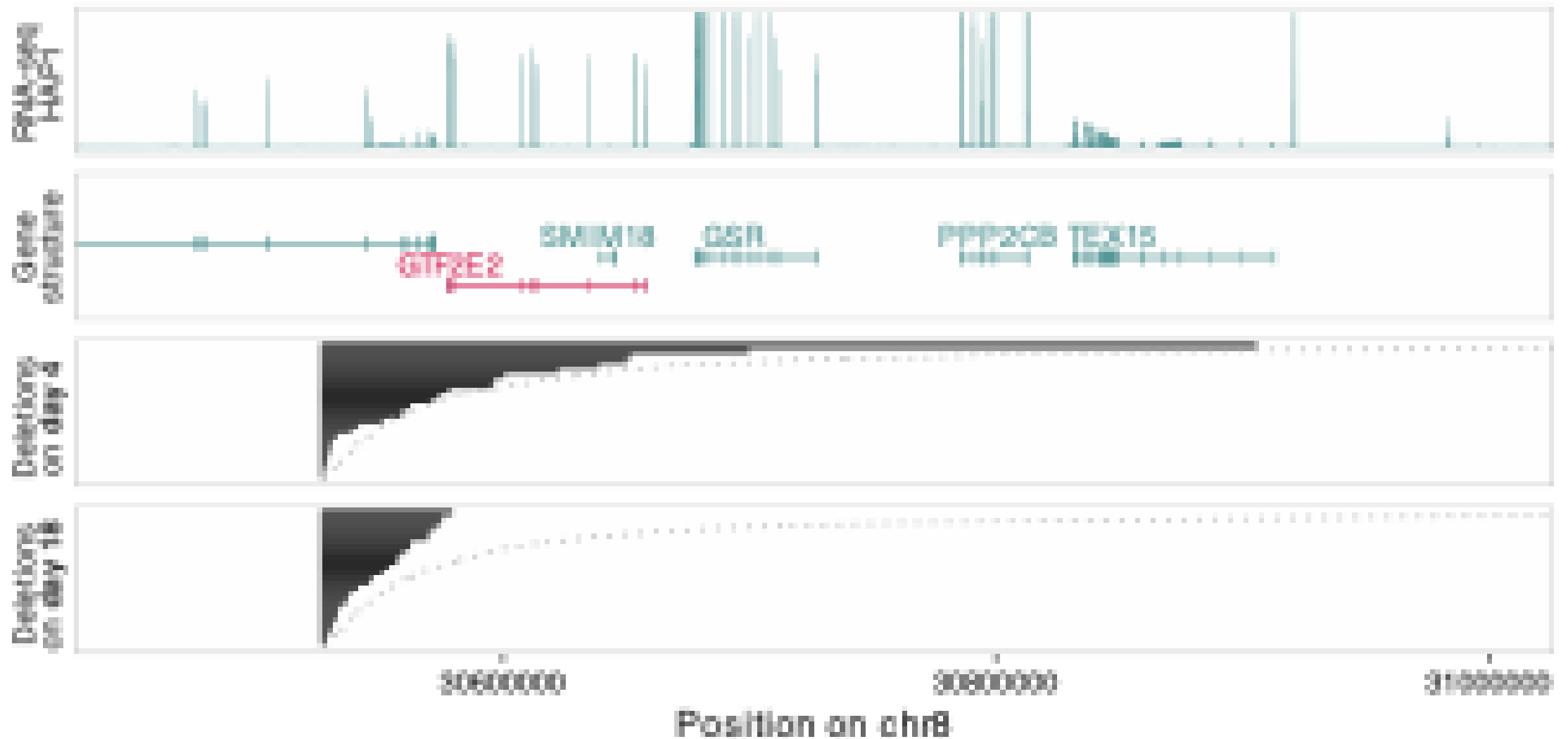
An evolutionary SV effect predictor separates experimental variants in haploid and diploid cells



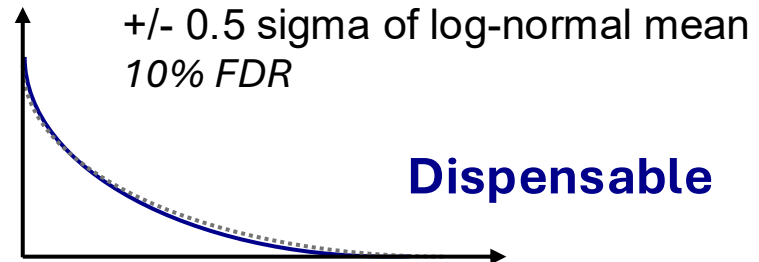
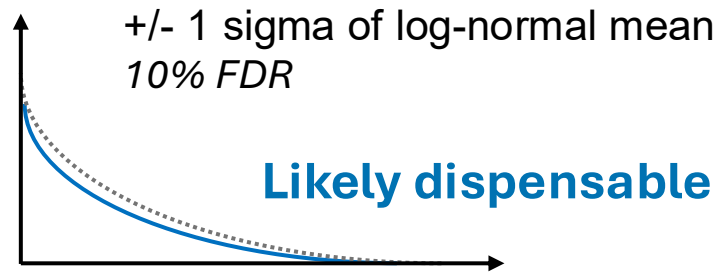
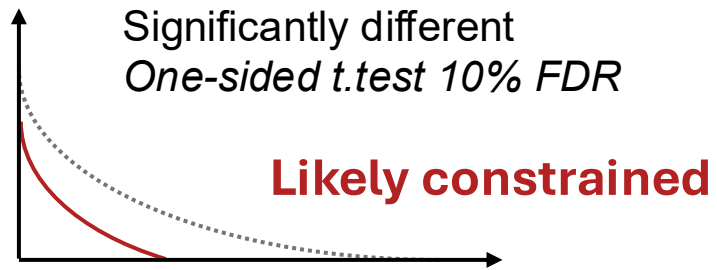
Example for a location not under selection



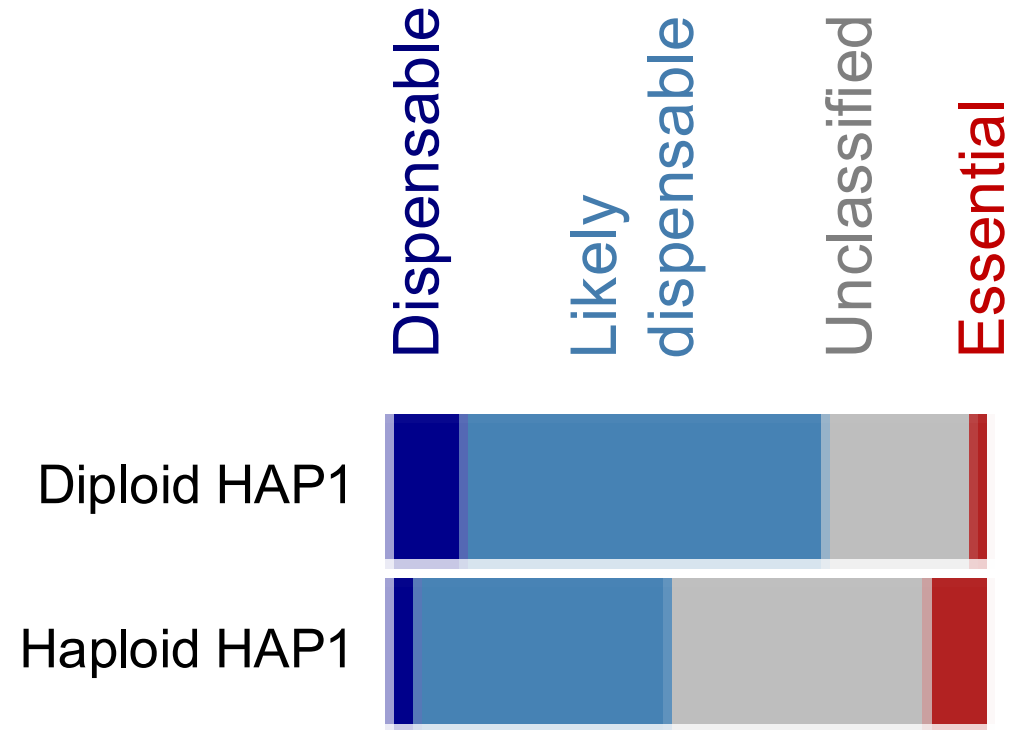
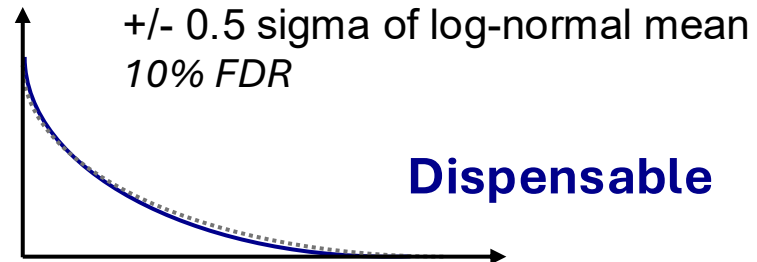
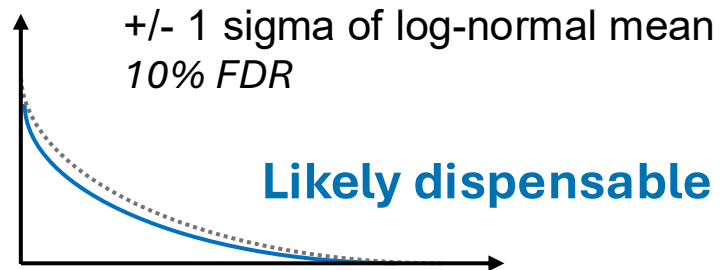
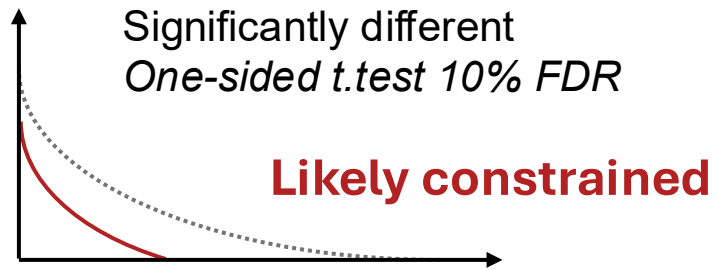
Example for a location under selection

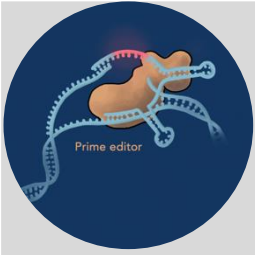


Can we estimate the necessity for all 30 kb intervals in the human genome?



Can we estimate the necessity for all 30 kb intervals in the human genome?

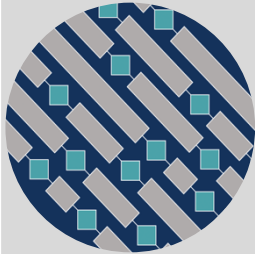




Optimizing genome editors
How do we optimally insert sequences?



Koeppel, Weller,
Peets et al., 2023



Engineering regulatory landscapes
How do enhancer clusters work?



Koeppel, Murat,
et al., 2025



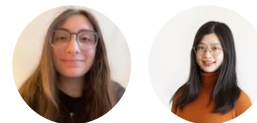
Scrambling entire human genomes
What is the fitness landscape of SVs?



Koeppel, Ferreira,
et al., 2025



High-resolution deletion scanning
How much of the genome is essential?



In preparation

Research program: Engineering SV to understand genome function

SV engineering before 2019

Recombinase chromosome engineering

- Requires pre-engineering of a landing pad
- Low throughput



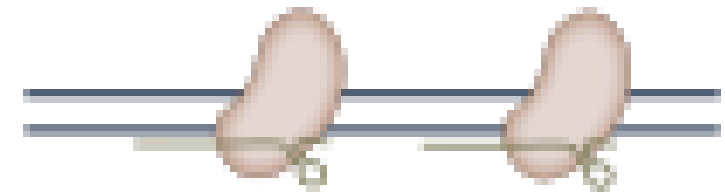
Transposon-based

- Largely restricted to insertions
- No control over SV junctions



Dual-cut with nuclease

- Heterogenous outcomes
- Low efficiency of on-target variants



The time for SV engineering is now!

Targeted DNA integration in human cells without double-strand breaks using CRISPR-associated transposases

George D. Lampe, Rebeca T. King, Tyler S. Halpin-Healy, Sanne E. Klompe, Marcus I. Hogan, Phuc Leo H. Vo, Stephen Tang, Alejandro Chavez & Samuel H. Sternberg ✉
Nature Biotechnology 42, 87–98 (2024) | [Cite this article](#)
31k Accesses | 113 Citations | 174 Altmetric | [Metrics](#)

Megabase-scale human genome rearrangement with programmable bridge recombinases

NICHOLAS T. PERRY, LIAM J. BARTIE, DHIRVJA KATREKAR, GABRIEL A. GONZALEZ, MATTHEW G. DURRANT, JAMES J. PAI, ALISON FANTON, JULIANA Q. MARTINS, MASAHIRO HIRAZUMI, L.-J. AND PATRICK D. HSU ✉
+3 authors [Authors Info & Affiliations](#)
SCIENCE • 25 Sep 2025 • First Release • DOI:10.1126/science.adr0276

Programmable gene insertion in human cells with a laboratory-evolved CRISPR-associated transposase

ISAAC P. WITTE, GEORGE D. LAMPE, SIMON EITZINGER, SHANNON M. MILLER, KARA N. BERRIOS, AMBER M. MCLEROY, REBECA T. KING, OLIVIA G. STRINGHAM, DIEGO R. GELSINGER, L.-J. AND DAVID R. LIU ✉
+5 authors [Authors Info & Affiliations](#)

Systematic discovery of recombinases for efficient integration of large DNA sequences into the human genome

Matthew G. Durrant, Alison Fanton, Josh Tycko, Michaela Hinks, Sita S. Chandrasekaran, Nicholas T. Perry, Julia Schaepe, Peter P. Du, Peter Lotfy, Michael C. Bassik, Lacramioara Bintu, Ami S. Bhatt ✉ & Patrick D. Hsu ✉
Nature Biotechnology 41, 488–499 (2023) | [Cite this article](#)
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Bridge RNAs direct programmable recombination of target and donor DNA

Matthew G. Durrant, Nicholas T. Perry, James J. Pai, Aditya R. Jangid, Januka S. Athukoralage, Masahiro Hiraizumi, John P. McSpedon, April Pawluk, Hiroshi Nishimasu, Silvana Konermann & Patrick D. Hsu ✉

Efficient site-specific integration of large genes in mammalian cells via continuously evolved recombinases and prime editing

Smriti Pandey, Xin D. Gao, Nicholas A. Krasnow, Amber McElroy, Y. Allen Tao, Jordyn E. Duby, Benjamin J. Steinbeck, Julia McCreary, Sarah E. Pierce, Jakub Tolar, Torsten B. Meissner, Elliot L. Chaikof, Mark J. Osborn & David R. Liu ✉

Site-specific DNA insertion into the human genome with engineered recombinases

Alison Fanton, Liam J. Bartie, Juliana Q. Martins, Vincent Q. Tran, Laine Goudy, Courtney Kernick, Matthew G. Durrant, Jingyi Wei, Zev Armour-Garb, April Pawluk, Silvana Konermann, Alexander Marson, Luke A. Gilbert, Theodore L. Roth & Patrick D. Hsu ✉
Nature Biotechnology (2025) | [Cite this article](#)

Precise genomic deletions using paired prime editing

Junhong Choi, Wei Chen, Chase C. Suiter, Choli Lee, Florence M. Chardon, Wei Yang, Anh Leith, Riza M. Daza, Beth Martin & Jay Shendure ✉
Nature Biotechnology 40, 218–226 (2022) | [Cite this article](#)

Article

Search-and-replace genome editing without double-strand breaks or donor DNA

<https://doi.org/10.1038/s41586-019-1711-4>
Received: 26 August 2019
Accepted: 10 October 2019

Andrew V. Anzalone^{1,2,3}, Peyton B. Randolph^{1,2,3}, Jessie R. Davis^{1,2,3}, Alexander A. Sousa^{1,2,3}, Luke W. Koblan^{1,2,3}, Jonathan M. Levy^{1,2,3}, Peter J. Chen^{1,2,3}, Christopher Wilson^{1,2,3}, Gregory A. Newby^{1,2,3}, Aditya Raguram^{1,2,3} & David R. Liu^{1,2,3*}

Systematic Development of Reprogrammed Modular Integrases Enables Precise Genomic Integration of Large DNA Sequences

Friedrich Fauser^{1,2,3}, Sebastian Arangundy-Franklin^{1,2}, Jessica E. Davis¹, Lifeng Liu¹, Nicola J. Schmidt¹, Luis Rodriguez¹, Nicholas A. Scarlott¹, Rakshaa Mureli¹, Danny F. Xia¹, Sarah J. Hinkley¹, Bhakti N. Kadam¹, Nga Nguyen¹, Stephen Lam¹, Bryan Bourgeois¹, Emily Tait¹, Mohammad Qasim¹, Vishvesha Vaidya¹, Adeline Chen¹, Andrew Nguyen¹, Patrick Li¹, David E. Paschon¹, Gregory D. David¹ & David R. Liu^{1,2,3*}

Iterative recombinase technologies for efficient and precise genome engineering across kilobase to megabase scales

Chao Sun^{1,3}, Hongchao Li^{1,3}, Yijing Liu^{1,2,3}, Yunjia Li^{1,2}, Rui Gao^{1,2}, Xiaoli Shi¹, Hongyuan Fei¹, Chao Sun^{1,3}, Hongchao Li^{1,3}, Yijing Liu^{1,2,3}, Yunjia Li^{1,2}, Rui Gao^{1,2}, Xiaoli Shi¹, Hongyuan Fei¹, Chao Sun^{1,3}, Hongchao Li^{1,3}, Yijing Liu^{1,2,3}, Yunjia Li^{1,2}, Rui Gao^{1,2}, Xiaoli Shi¹, Hongyuan Fei¹

Drag-and-drop genome insertion of large sequences without double-strand DNA cleavage using CRISPR-directed integrases

Matthew T. N. Yarnall, Eleonora I. Ioannidi, Cian Schmitt-Ulms, Rohan N. Krajeski, Justin Lim, Lukas Villiger, Wenyuan Zhou, Kaiyi Jiang, Sofya K. Garushyants, Nathaniel Roberts, Liyang Zhang, Christopher A. Vakulskas, John A. Walker II, Anastasia P. Kadina, Adrianna E. Zepeda, Kevin Holden, Hong Ma, Jun Xie, Guangping Gao, Lander Foquet, Greg Bial, Sara K. Donnelly, Yoshinari Miyata, R. Radloff

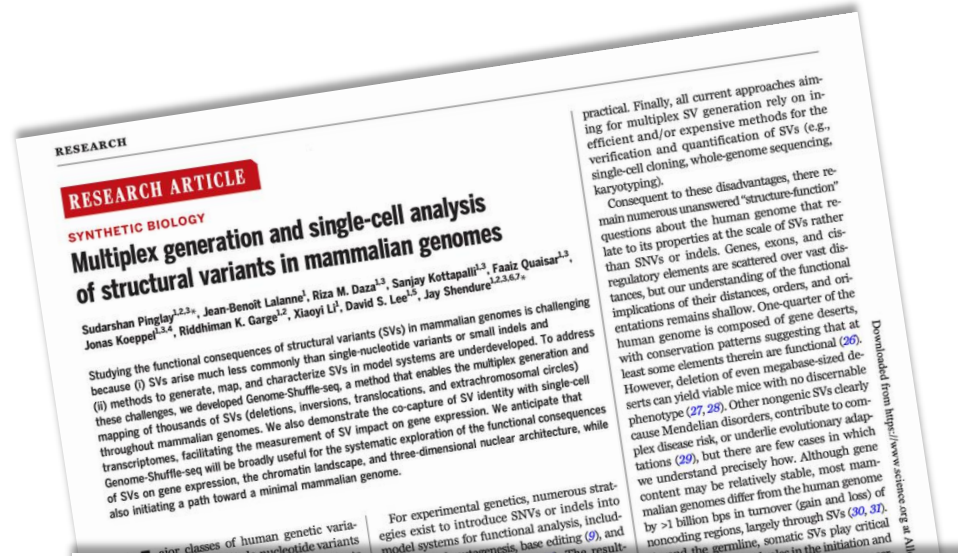
Programmable deletion, replacement, integration and inversion of large DNA sequences with twin prime editing

Andrew V. Anzalone, Xin D. Gao, Christopher J. Podracky, Andrew T. Nelson, Luke W. Koblan, Aditya M. Levy, Jaron A. M. Mercer & David R. Liu ✉

Efficient site-specific integration of large genes in mammalian cells via continuously evolved recombinases and prime editing

Smriti Pandey, Xin D. Gao, Nicholas A. Krasnow, Amber McElroy, Y. Allen Tao, Jordyn E. Duby, Benjamin J. Steinbeck, Julia McCreary, Sarah E. Pierce, Jakub Tolar, Torsten B. Meissner, Elliot L. Chaikof, Mark J. Osborn & David R. Liu ✉
Nature Biomedical Engineering 9, 22–39 (2025) | [Cite this article](#)

The potential for large-scale SV engineering is largely untapped!



ARTICLES

nature
biotechnology

Complete humanization of the mouse immunoglobulin loci enables efficient therapeutic antibody discovery

E-Chiang Lee¹, Qi Liang¹, Hanif Ali¹, Luke Bayliss¹, Alastair Beasley¹, Tara Bloomfield-Gerdes¹, Laura Bonoli¹, Richard Brown¹, Jamie Campbell¹, Adam Carpenter¹, Sara Chalk¹, Alison Davis¹, Nick England¹, Alla Fane-Dremucheva¹, Bettina Franz¹, Volker Germaschewski¹, Helen Holmes¹, Steve Holmes¹, Ian Kirby¹, Miha Kosmac¹, Anais Legent¹, Hui Lui¹, Anais Manin¹, Siobhan O'Leary¹, Jemima Paterson¹, Rocco Sciarillo¹, Anneliese Speak², Dominik Spensberger¹, Laura Tuffery¹, Nikole Waddell¹, Wei Wang², Sophie Wells¹, Vivian Wong¹, Andrew Wood¹, Michael J Owen¹, Glenn A Friedrich¹ & Allan Bradley^{1,2}

If immunized with an antigen of interest, transgenic mice with large portions of unrearranged human immunoglobulin loci can produce fully human antigen-specific antibodies; several such antibodies are in clinical use. However, technical limitations inherent to conventional transgenic technology and sequence divergence between the human and mouse



3456-3474 Nucleic Acids Research, 2022, Vol. 50, No. 6
<https://doi.org/10.1093/nar/gkac153>

Published online 4 March 2022

Global chromosome rearrangement induced by CRISPR-Cas9 reshapes the genome and transcriptome of human cells

Ying Liu^{1,†}, Guangwei Ma^{1,2,†}, Zenghong Gao¹, Jian Li¹, Jin Wang¹, Xiangping Zhu¹, Ruowu Ma¹, Jiawen Yang^{1,3}, Yiting Zhou¹, Kaishun Hu¹, Yin Zhang^{1,*} and Yabin Guo^{1,*}

¹Guangdong Provincial Key Laboratory of Malignant Tumor Epigenetics and Gene Regulation, Guangdong-Hong Kong Joint Laboratory for RNA Medicine, Medical Research Center, Sun Yat-sen Memorial Hospital, Sun Yat-sen University, Guangzhou 510120, China, ²Ministry of Education Key Laboratory for Ecology of Tropical Islands, Key Laboratory of Tropical Animal and Plant Ecology of Hainan Province, College of Life Sciences, Hainan Normal University, Haikou 571158, China and ³State Key Laboratory of Biocatalysis, School of Life Sciences, Sun Yat-sen University, Guangzhou 510275, China

Received July 08, 2021; Revised January 29, 2022; Editorial Decision February 17, 2022; Accepted February 19, 2022

ABSTRACT

Chromosome rearrangement plays important roles in development, carcinogenesis, and

INTRODUCTION

Chromosome rearrangement

Why is it untapped? Very few labs have all the ingredients!



Read out strategies

- High-throughput
- Capture junctions directly
- Multimodal phenotyping

Getting editors working reliably

- Novel editors
- Surprising off-target activity
- Unexplored guide/experiment design space
- Burden on cells



Making sense of the data

- High density of expertise on genomes

Research program aims

1. Build an SV screening platform
2. Use it to systematically delete genome features
3. Make highly rearranged/minimized genomes

Building an SV screening platform with multimodal readout

Design criteria

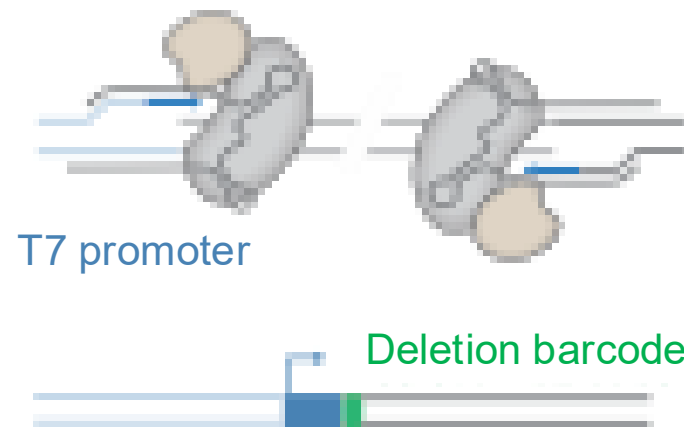
Read SV junctions directly

High throughput

Compatible with scRNA-seq

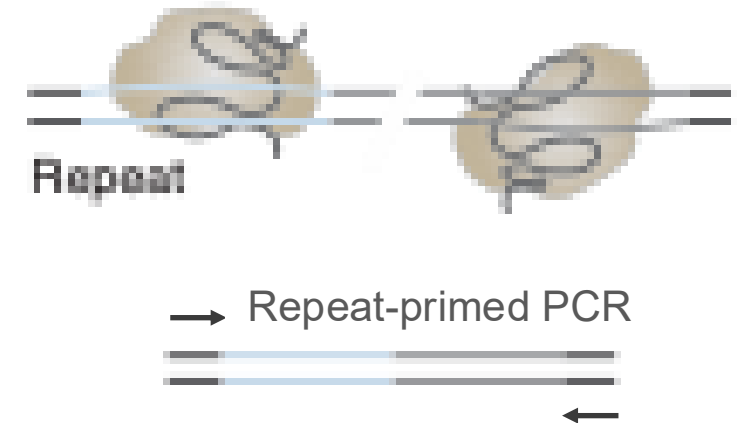
500bp – 1 kb scale

Prime deletion



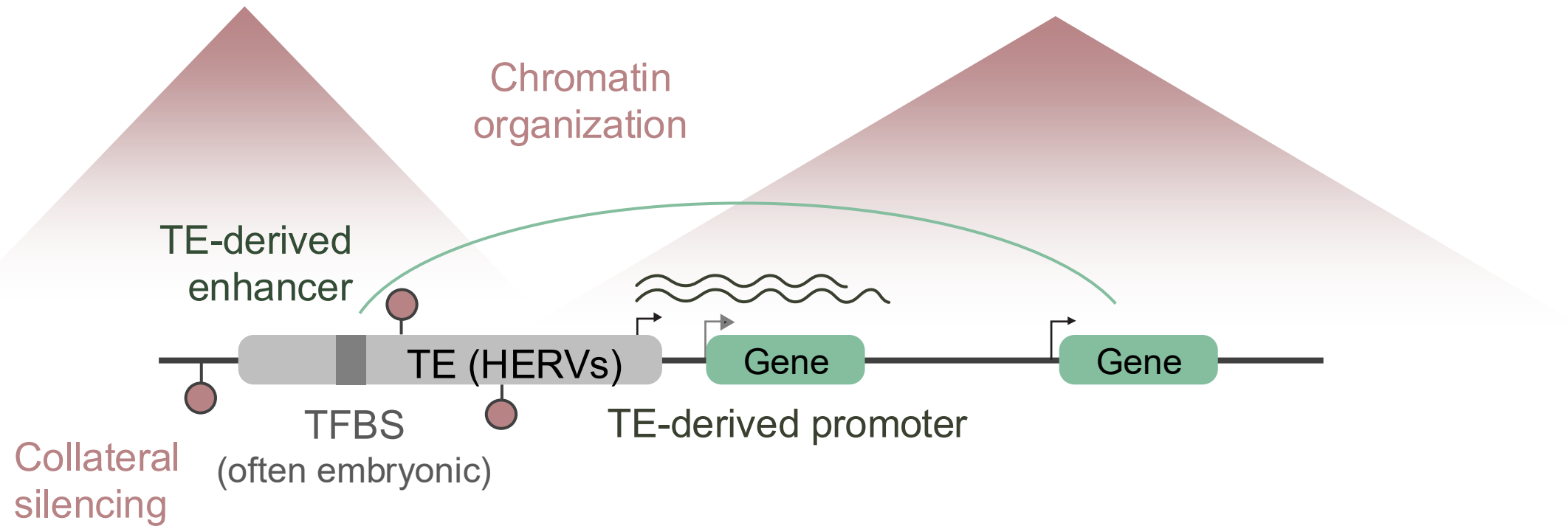
100 kb + scale

Bridge recombinase



Transposable elements are an excellent screening candidate

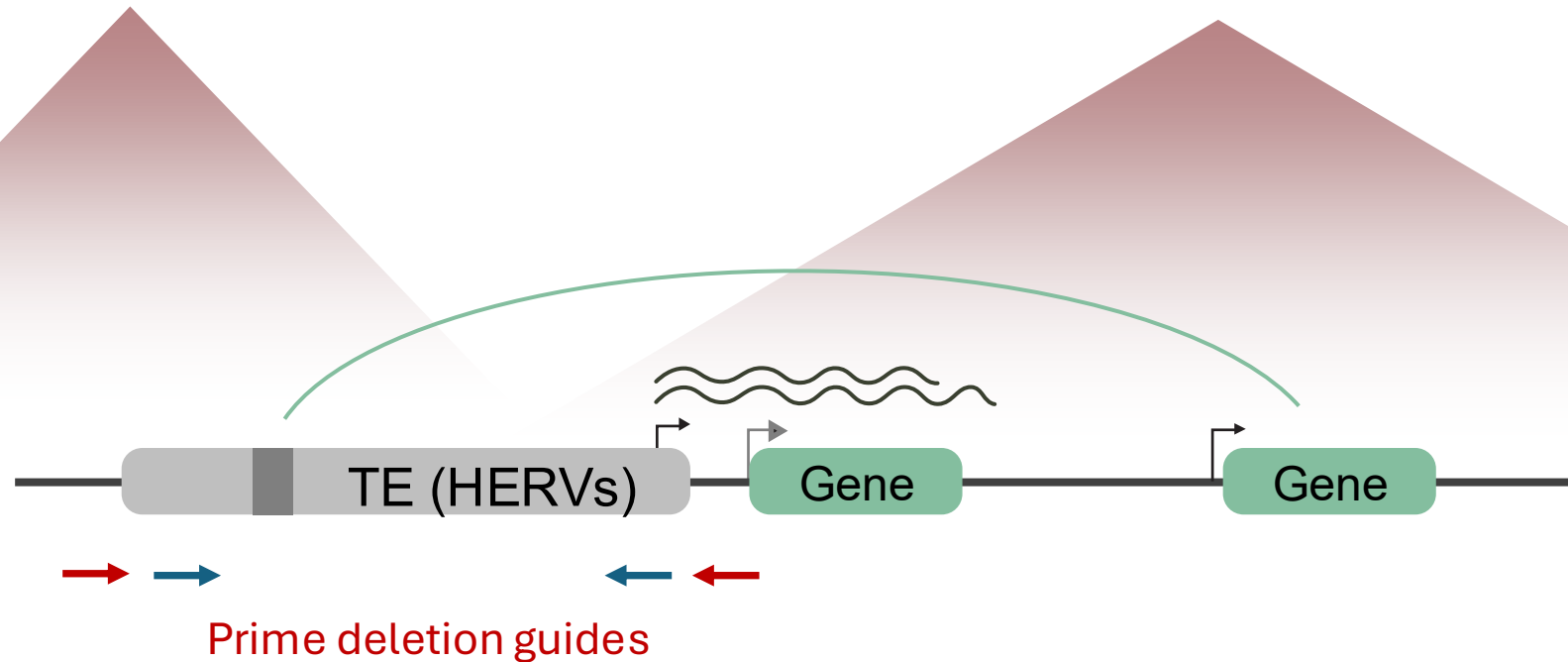
What fraction has a function in cells?
How should we design a minimal genome?



Difficult to probe with traditional experimental perturbation strategies

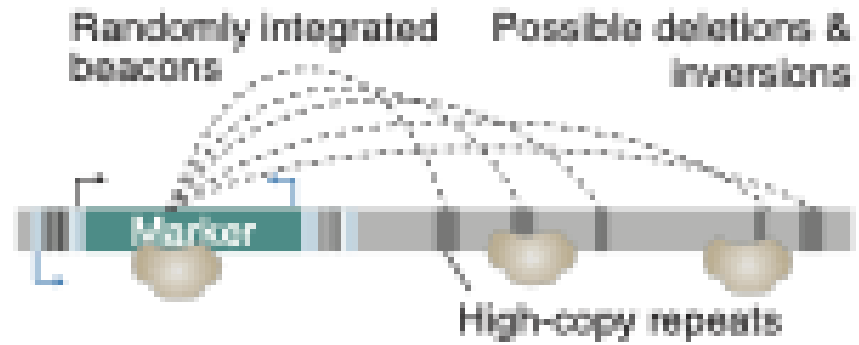
Transposable elements are an excellent screening candidate

What fraction has a function in cells?
How should we design a minimal genome?



Difficult to probe with traditional experimental perturbation strategies

Minimizing genomes to unveil redundancies



The road ahead

Direct outputs

Optimized SV editors

Genome-wide SV
screens

Minimized genomes

SV-phenotypes
atlases

Applications

Repair pathogenic SVs

Functional annotation of
non-coding DNA

Synthetic-lethal targets for
therapy

SV pathogenicity prediction

Long-term vision

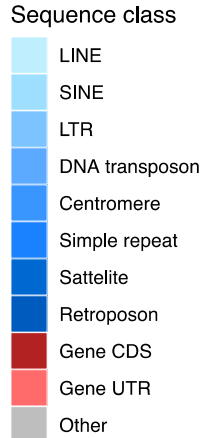
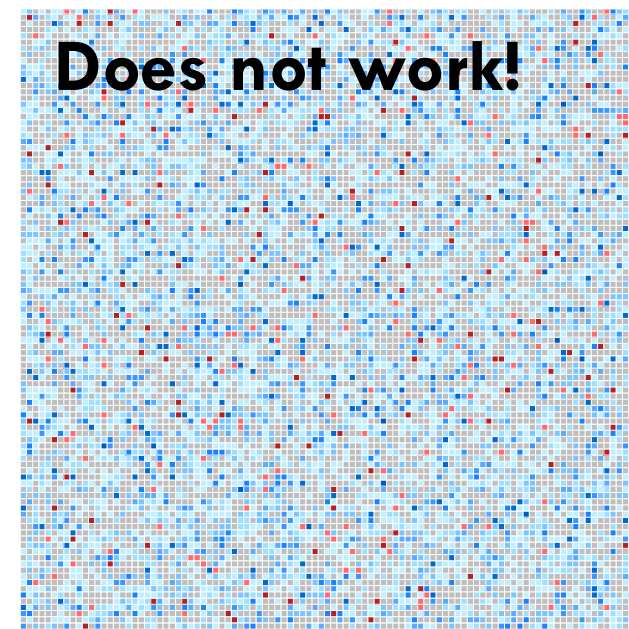
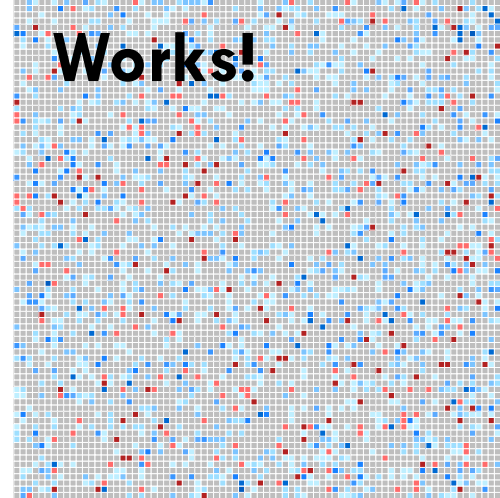
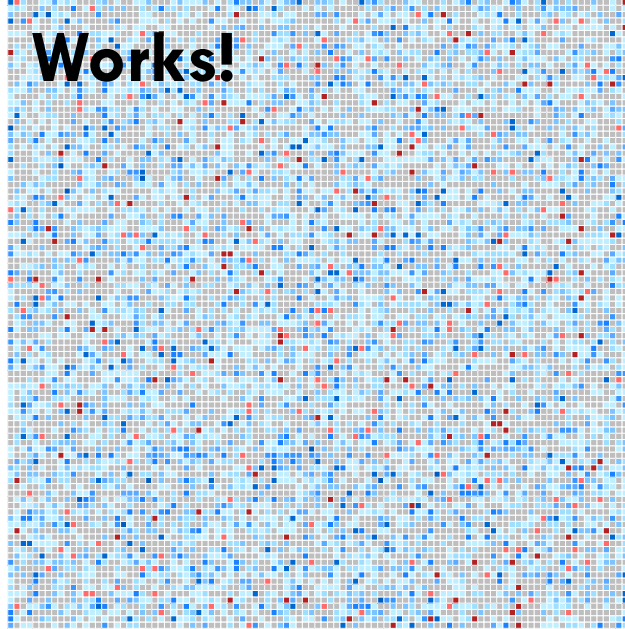
Large-scale genome engineering as
routine as gene knockouts

Predictive understanding of genome
architecture

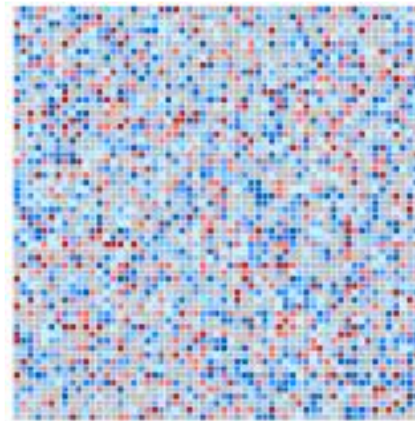
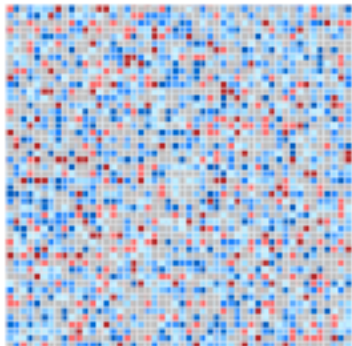
Minimal genomes built to specification



Towards predictive understanding of genome architecture



Can only divide



**Can divide and
differentiate into muscle**

Acknowledgements I

Shendure lab

Sudarshan Pinglay
Jay Shendure
Faaiz Quaisar
Sam Sgrizzi
Eleftheria Beres
Carina Biar
Peixi Chen
Riza Daza
Hanna Liao
Xiaoyi Li
Abby McGee
Olga Oseth
Molly Gasperini
Aiden Keith



Yi Fu
Mohammed Aljohani
Zach Amador
Zihao Song
Jordan Knuth
Nahum Smith
Jesse Gray

Alessandro Coradini
Elin Madli Peets
Flo Chardon
Shannon Smith
John Atwater
David Mai
Renke Tan

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Shruthi Vijay Kumar
Kenny Dang
Mary O'Neill
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Yan Zhang

Pinglay lab



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Seattle Hub
for **SYNTHETIC BIOLOGY**

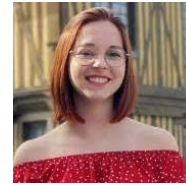
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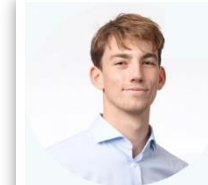
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Fabio Liberante



Elin Peets



Raphael Ferreira



Ronnie Crawford



Tom Ellis



Klaudia Ciurkot

Vivek Iyer, Gareth Girling, Thomas Vanderstichele, Jacob Hepkema, Ananth Pallaseni, Megan Gozzard, Jannat Ijaz, Peter Campbell, Allan Bradley



And everyone else in the Parts lab!



@JonasKoeppel @LeopoldParts



jk24@sanger.ac.uk lp2@sanger.ac.uk

